

Mathematics: analysis and approaches guide

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Diploma Programme

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IB mission statement

The International Baccalaureate aims to develop inquiring, knowledgeable and caring young people who help to create a better and more peaceful world through intercultural understanding and respect. To this end the organization works with schools, governments and international organizations to develop challenging programmes of international education and rigorous assessment. These programmes encourage students across the world to become active, compassionate and lifelong learners who understand that other people, with their differences, can also be right.



IB learner profile

The aim of all IB programmes is to develop internationally minded people who, recognizing their common humanity and shared guardianship of the planet, help to create a better and more peaceful world.

As IB learners we strive to be:

INQUIRERS

We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.

KNOWLEDGEABLE

We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.

THINKERS

We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions.

COMMUNICATORS

We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.

PRINCIPLED

We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.

OPEN-MINDED

We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.

CARING

We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.

RISK-TAKERS

We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.

BALANCED

We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.

REFLECTIVE

We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.

The IB learner profile represents 10 attributes valued by IB World Schools. We believe these attributes, and others like them, can help individuals and groups become responsible members of local, national and global communities.

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Purpose of this document

This publication is intended to guide the planning, teaching and assessment of the subject in schools. Subject teachers are the primary audience, although it is expected that teachers will use the guide to inform students and parents about the subject.

This guide can be found on the subject page of the programme resource centre at resources.ibo.org, a password-protected IB website designed to support IB teachers. It can also be purchased from the IB store at store.ibo.org

Additional resources

Additional publications such as specimen papers and markschemes, teacher support materials, subject reports and grade descriptors can also be found on the programme resource centre. Past examination papers as well as markschemes can be purchased from the IB store.

Teachers are encouraged to check the programme resource centre for additional resources created or used by other teachers. Teachers can provide details of useful resources, for example websites, books, videos, journals or teaching ideas.

Acknowledgment

The IB wishes to thank the educators and associated schools for generously contributing time and resources to the production of this guide.

First assessment 2021

The Diploma Programme

The Diploma Programme is a rigorous pre-university course of study designed for students in the 16-19 age range. It is a broad-based two-year course that aims to encourage students to be knowledgeable and inquiring, but also caring and compassionate. There is a strong emphasis on encouraging students to develop intercultural understanding, open-mindedness, and the attitudes necessary for them to respect and evaluate a range of points of view.

The Diploma Programme model

The course is presented as six academic areas enclosing a central core (see figure 1). It encourages the concurrent study of a broad range of academic areas. Students study two modern languages (or a modern language and a classical language), a humanities or social science subject, an experimental science, mathematics and one of the creative arts. It is this comprehensive range of subjects that makes the Diploma Programme a demanding course of study designed to prepare students effectively for university entrance. In each of the academic areas students have flexibility in making their choices, which means they can choose subjects that particularly interest them and that they may wish to study further at university.

Figure 1

The Diploma Programme model



Choosing the right combination

Students are required to choose one subject from each of the six academic areas, although they can, instead of an arts subject, choose two subjects from another area. Normally, three subjects (and not more than four) are taken at higher level (HL), and the others are taken at standard level (SL). The IB recommends 240 teaching hours for HL subjects and 150 hours for SL. Subjects at HL are studied in greater depth and breadth than at SL.

At both levels many skills are developed, especially those of critical thinking and analysis. At the end of the course, students' abilities are measured by means of external assessment. Many subjects contain some element of coursework assessed by teachers.

The core of the Diploma Programme model

All Diploma Programme (DP) students participate in the three course elements that make up the core of the model.

Theory of knowledge (TOK) is a course that is fundamentally about critical thinking and inquiry into the process of knowing rather than about learning a specific body of knowledge. The TOK course examines the nature of knowledge and how we know what we claim to know. It does this by encouraging students to analyse knowledge claims and explore questions about the construction of knowledge. The task of TOK is to emphasize connections between areas of shared knowledge and link them to personal knowledge in such a way that an individual becomes more aware of his or her own perspectives and how they might differ from others.

Creativity, activity, service (CAS) is at the heart of the Diploma Programme. The emphasis in CAS is on helping students to develop their own identities, in accordance with the ethical principles embodied in the IB mission statement and the IB learner profile. It involves students in a range of activities alongside their academic studies throughout the Diploma Programme. The three strands of CAS are creativity (arts, and other experiences that involve creative thinking), activity (physical exertion contributing to a healthy lifestyle) and service (an unpaid and voluntary exchange that has a learning benefit for the student). Possibly, more than any other component in the Diploma Programme, CAS contributes to the IB's mission to create a better and more peaceful world through intercultural understanding and respect.

The extended essay, including the world studies extended essay, offers the opportunity for IB students to investigate a topic of special interest, in the form of a 4,000-word piece of independent research. The area of research undertaken is chosen from one of the students' six Diploma Programme subjects, or in the case of the inter-disciplinary World Studies essay, two subjects, and acquaints them with the independent research and writing skills expected at university. This leads to a major piece of formally-presented, structured writing, in which ideas and findings are communicated in a reasoned and coherent manner, appropriate to the subject or subjects chosen. It is intended to promote high-level research and writing skills, intellectual discovery and creativity. An authentic learning experience, it provides students with an opportunity to engage in personal research on a topic of choice, under the guidance of a supervisor.

Approaches to teaching and approaches to learning

Approaches to teaching and learning across the Diploma Programme refers to deliberate strategies, skills and attitudes which permeate the teaching and learning environment. These approaches and tools, intrinsically linked with the learner profile attributes, enhance student learning and assist student preparation for the Diploma Programme assessment and beyond. The aims of approaches to teaching and learning in the Diploma Programme are to:

- empower teachers as teachers of learners as well as teachers of content
- empower teachers to create clearer strategies for facilitating learning experiences in which students are more meaningfully engaged in structured inquiry and greater critical and creative thinking

- promote both the aims of individual subjects (making them more than course aspirations) and linking previously isolated knowledge (concurrency of learning)
- encourage students to develop an explicit variety of skills that will equip them to continue to be actively engaged in learning after they leave school, and to help them not only obtain university admission through better grades but also prepare for success during tertiary education and beyond
- enhance further the coherence and relevance of the students' Diploma Programme experience
- allow schools to identify the distinctive nature of an IB Diploma Programme education, with its blend of idealism and practicality.

The five approaches to learning (developing thinking skills, social skills, communication skills, self-management skills and research skills) along with the six approaches to teaching (teaching that is inquiry-based, conceptually-focused, contextualized, collaborative, differentiated and informed by assessment) encompass the key values and principles that underpin IB pedagogy.

The IB mission statement and the IB learner profile

The Diploma Programme aims to develop in students the knowledge, skills and attitudes they will need to fulfill the aims of the IB, as expressed in the organization's mission statement and the learner profile. Teaching and learning in the Diploma Programme represent the reality in daily practice of the organization's educational philosophy.

Academic honesty

Academic honesty in the Diploma Programme is a set of values and behaviours informed by the attributes of the learner profile. In teaching, learning and assessment, academic honesty serves to promote personal integrity, engender respect for the integrity of others and their work, and ensure that all students have an equal opportunity to demonstrate the knowledge and skills they acquire during their studies.

All coursework—including work submitted for assessment—is to be authentic, based on the student's individual and original ideas with the ideas and work of others fully acknowledged. Assessment tasks that require teachers to provide guidance to students or that require students to work collaboratively must be completed in full compliance with the detailed guidelines provided by the IB for the relevant subjects.

For further information on academic honesty in the IB and the Diploma Programme, please consult the IB publications *Academic honesty in the IB educational context*, *Effective citing and referencing*, *The Diploma Programme: From principles into practice* and *General regulations: Diploma Programme*. Specific information regarding academic honesty as it pertains to external and internal assessment components of this Diploma Programme subject can be found in this guide.

Acknowledging the ideas or work of another person

Coordinators and teachers are reminded that candidates must acknowledge all sources used in work submitted for assessment. The following is intended as a clarification of this requirement.

Diploma Programme candidates submit work for assessment in a variety of media that may include audio-visual material, text, graphs, images and/or data published in print or electronic sources. If a candidate uses the work or ideas of another person, the candidate must acknowledge the source using a standard style of referencing in a consistent manner. A candidate's failure to acknowledge a source will be investigated by the IB as a potential breach of regulations that may result in a penalty imposed by the IB final award committee.

The IB does not prescribe which style(s) of referencing or in-text citation should be used by candidates; this is left to the discretion of appropriate faculty/staff in the candidate's school. The wide range of subjects, three response languages and the diversity of referencing styles make it impractical and restrictive to insist on particular styles. In practice, certain styles may prove most commonly used, but schools are free to

choose a style that is appropriate for the subject concerned and the language in which candidates' work is written. Regardless of the reference style adopted by the school for a given subject, it is expected that the minimum information given includes: name of author, date of publication, title of source, and page numbers as applicable.

Candidates are expected to use a standard style and use it consistently so that credit is given to all sources used, including sources that have been paraphrased or summarized. When writing text, candidates must clearly distinguish between their words and those of others by the use of quotation marks (or other method, such as indentation) followed by an appropriate citation that denotes an entry in the bibliography. If an electronic source is cited, the date of access must be indicated. Candidates are not expected to show faultless expertise in referencing, but are expected to demonstrate that all sources have been acknowledged. Candidates must be advised that audio-visual material, text, graphs, images and/or data published in print or in electronic sources that is not their own must also attribute the source. Again, an appropriate style of referencing/citation must be used.

Learning diversity and learning support requirements

Schools must ensure that equal access arrangements and reasonable adjustments are provided to candidates with learning support requirements that are in line with the IB documents *Access and inclusion policy* and *Learning diversity and inclusion in IB programmes*.

The documents *Meeting student learning diversity in the classroom* and *The IB guide to inclusive education: a resource for whole school development* are available to support schools in the ongoing process of increasing access and engagement by removing barriers to learning.

Nature of Mathematics

Introduction

Mathematics has been described as the study of structure, order and relation that has evolved from the practices of counting, measuring and describing objects. Mathematics provides a unique language to describe, explore and communicate the nature of the world we live in as well as being a constantly building body of knowledge and truth in itself that is distinctive in its certainty. These two aspects of mathematics, a discipline that is studied for its intrinsic pleasure and a means to explore and understand the world we live in, are both separate yet closely linked.

Mathematics is driven by abstract concepts and generalization. This mathematics is drawn out of ideas, and develops through linking these ideas and developing new ones. These mathematical ideas may have no immediate practical application. Doing such mathematics is about digging deeper to increase mathematical knowledge and truth. The new knowledge is presented in the form of theorems that have been built from axioms and logical mathematical arguments and a theorem is only accepted as true when it has been proven. The body of knowledge that makes up mathematics is not fixed; it has grown during human history and is growing at an increasing rate.

The side of mathematics that is based on describing our world and solving practical problems is often carried out in the context of another area of study. Mathematics is used in a diverse range of disciplines as both a language and a tool to explore the universe; alongside this its applications include analyzing trends, making predictions, quantifying risk, exploring relationships and interdependence.

While these two different facets of mathematics may seem separate, they are often deeply connected. When mathematics is developed, history has taught us that a seemingly obscure, abstract mathematical theorem or fact may in time be highly significant. On the other hand, much mathematics is developed in response to the needs of other disciplines.

The two mathematics courses available to Diploma Programme (DP) students express both the differences that exist in mathematics described above and the connections between them. These two courses might approach mathematics from different perspectives, but they are connected by the same mathematical body of knowledge, ways of thinking and approaches to problems. The differences in the courses may also be related to the types of tools, for instance technology, that are used to solve abstract or practical problems. The next section will describe in more detail the two available courses.

Summary of courses available

Individual students have different needs, aspirations, interests and abilities. For this reason there are two different subjects in mathematics, each available at SL and HL. These courses are designed for different types of students: those who wish to study mathematics as a subject in its own right or to pursue their interests in areas related to mathematics, and those who wish to gain understanding and competence in how mathematics relates to the real world and to other subjects. Each course is designed to meet the needs of a particular group of students. Mathematics: analysis and approaches and Mathematics: applications and interpretation are both offered at SL and HL. Therefore, great care should be taken to select the course and level that is most appropriate for an individual student.

In making this selection, individual students should be advised to take into account the following factors:

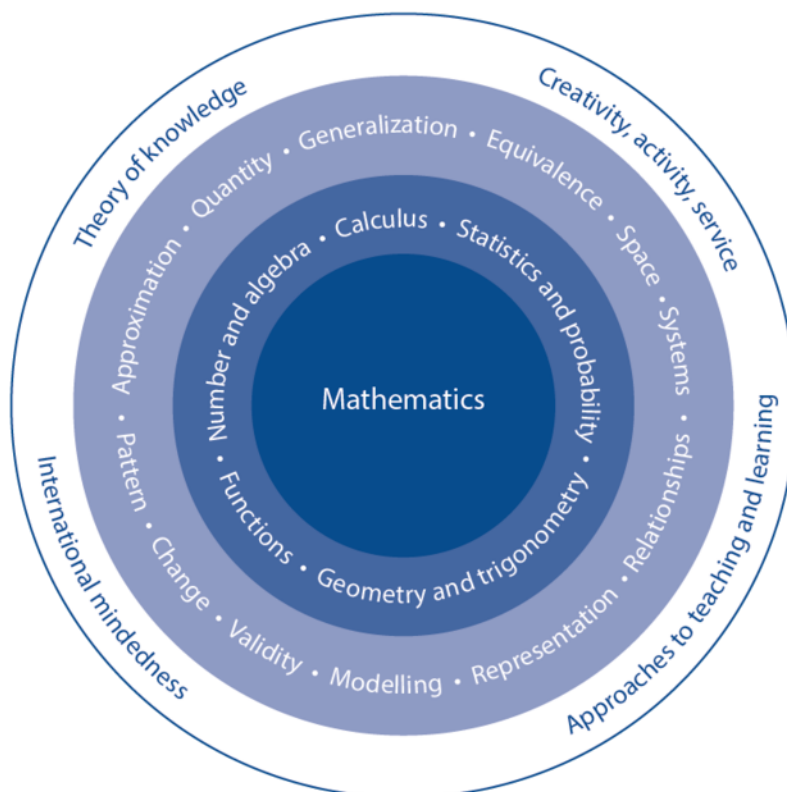
- their own abilities in mathematics and the type of mathematics in which they can be successful
- their own interest in mathematics and those particular areas of the subject that may hold the most interest for them
- their other choices of subjects within the framework of the DP or Career-related Programme (CP)

- their academic plans, in particular the subjects they wish to study in the future
- their choice of career.

Teachers are expected to assist with the selection process and to offer advice to students.

The nature of IB mathematics courses

Figure 2
The mathematics model



The structure of IB DP mathematics courses, with two different routes to choose from, recognizes the two different aspects of mathematics discussed in the introduction.

Mathematics: analysis and approaches is for students who enjoy developing their mathematics to become fluent in the construction of mathematical arguments and develop strong skills in mathematical thinking. They will also be fascinated by exploring real and abstract applications of these ideas, with and without technology. Students who take **Mathematics: analysis and approaches** will be those who enjoy the thrill of mathematical problem solving and generalization.

Mathematics: applications and interpretation is for students who are interested in developing their mathematics for describing our world and solving practical problems. They will also be interested in harnessing the power of technology alongside exploring mathematical models. Students who take **Mathematics: applications and interpretation** will be those who enjoy mathematics best when seen in a practical context.

Both subjects are offered at HL and SL. There are many elements common to both subjects although the approaches may be different. Both subjects will prepare students with the mathematics needed for a range of further educational courses corresponding to the two approaches to mathematics set out above.

Mathematics: analysis and approaches

This course recognizes the need for analytical expertise in a world where innovation is increasingly dependent on a deep understanding of mathematics. This course includes topics that are both traditionally part of a pre-university mathematics course (for example, functions, trigonometry, calculus) as well as topics that are amenable to investigation, conjecture and proof, for instance the study of sequences and series at both SL and HL, and proof by induction at HL.

The course allows the use of technology, as fluency in relevant mathematical software and hand-held technology is important regardless of choice of course. However, Mathematics: analysis and approaches has a strong emphasis on the ability to construct, communicate and justify correct mathematical arguments.

Mathematics: analysis and approaches: Distinction between SL and HL

Students who choose Mathematics: analysis and approaches at SL or HL should be comfortable in the manipulation of algebraic expressions and enjoy the recognition of patterns and understand the mathematical generalization of these patterns. Students who wish to take Mathematics: analysis and approaches at higher level will have strong algebraic skills and the ability to understand simple proof. They will be students who enjoy spending time with problems and get pleasure and satisfaction from solving challenging problems.

Mathematics: applications and interpretation

This course recognizes the increasing role that mathematics and technology play in a diverse range of fields in a data-rich world. As such, it emphasizes the meaning of mathematics in context by focusing on topics that are often used as applications or in mathematical modelling. To give this understanding a firm base, this course also includes topics that are traditionally part of a pre-university mathematics course such as calculus and statistics.

The course makes extensive use of technology to allow students to explore and construct mathematical models. Mathematics: applications and interpretation will develop mathematical thinking, often in the context of a practical problem and using technology to justify conjectures.

Mathematics: applications and interpretation: Distinction between SL and HL

Students who choose Mathematics: applications and interpretation at SL or HL should enjoy seeing mathematics used in real-world contexts and to solve real-world problems. Students who wish to take Mathematics: applications and interpretation at higher level will have good algebraic skills and experience of solving real-world problems. They will be students who get pleasure and satisfaction when exploring challenging problems and who are comfortable to undertake this exploration using technology.

Mathematics and theory of knowledge

The relationship between each subject and theory of knowledge (TOK) is important and fundamental to the DP. The theory of knowledge course provides an opportunity for students to reflect on questions about how knowledge is produced and shared, both in mathematics and also across different areas of knowledge. It encourages students to reflect on their assumptions and biases, helping them to become more aware of their own perspective and the perspectives of others and to become “inquiring, knowledgeable and caring young people” (IB mission statement).

As part of their theory of knowledge course, students are encouraged to explore tensions relating to knowledge in mathematics. As an area of knowledge, mathematics seems to supply a certainty perhaps impossible in other disciplines and in many instances provides us with tools to debate these certainties. This may be related to the “purity” of the subject, something that can sometimes make it seem divorced

from reality. Yet mathematics has also provided important knowledge about the world and the use of mathematics in science and technology has been one of the driving forces for scientific advances.

Despite all its undoubted power for understanding and change, mathematics is in the end a puzzling phenomenon. A fundamental question for all knowers is whether mathematical knowledge really exists independently of our thinking about it. Is it there, “waiting to be discovered”, or is it a human creation? Indeed, the philosophy of mathematics is an area of study in its own right.

Students’ attention should be drawn to questions relating theory of knowledge (TOK) and mathematics, and they should also be encouraged to raise such questions themselves in both their mathematics and TOK classes. Examples of issues relating to TOK are given in the “Connections” sections of the syllabus. Further suggestions for making links to TOK can also be found in the mathematics section of the *Theory of knowledge guide*.

Mathematics and international-mindedness

International-mindedness is a complex and multi-faceted concept that refers to a way of thinking, being and acting characterized by an openness to the world and a recognition of our deep interconnectedness to others.

Despite recent advances in the development of information and communication technologies, the global exchange of mathematical information and ideas is not a new phenomenon and has been essential to the progress of mathematics. Indeed, many of the foundations of modern mathematics were laid many centuries ago by diverse civilisations – Arabic, Greek, Indian and Chinese among others.

Mathematics can in some ways be seen as an international language and, apart from slightly differing notation, mathematicians from around the world can communicate effectively within their field. Mathematics can transcend politics, religion and nationality, and throughout history great civilizations have owed their success in part to their mathematicians being able to create and maintain complex social and architectural structures. Politics has dominated the development of mathematics, to develop ballistics, navigation and trade, and land ownership, often influenced by governments and leaders. Many early mathematicians were political and military advisers and today mathematicians are integral members of teams who advise governments on where money and resources should be allocated.

Science and technology are of significant importance in today’s world. As the language of science, mathematics is an essential component of most technological innovation and underpins developments in science and technology, although the contribution of mathematics may not always be visible. Examples of this include the role of the binary number system, matrix algebra, network theory and probability theory in the digital revolution, or the use of mathematical simulations to predict future climate change or spread of disease. These examples highlight the key role mathematics can play in transforming the world around us.

One way of fostering international-mindedness is to provide opportunities for inquiry into a range of local and global issues and ideas. Many international organisations and bodies now exist to promote mathematics, and students are encouraged to access the resources and often-extensive websites of such mathematical organisations. This can enhance their appreciation of the international dimension of mathematics, as well as providing opportunities to engage with global issues surrounding the subject.

Examples of links relating to international-mindedness are given in the “Connections” sections of the syllabus.

Mathematics and creativity, activity, service (CAS)

CAS experiences can be associated with each of the subject groups of the DP.

CAS and mathematics can complement each other in a number of ways. Mathematical knowledge provides an important key to understanding the world in which we live, and the mathematical skills and techniques students learn in the mathematics courses will allow them to evaluate the world around them which will help them to develop, plan and deliver CAS experiences or projects.

An important aspect of the mathematics courses is that students develop the ability to systematically analyse situations and can recognize the impact that mathematics can have on the world around them. An

awareness of how mathematics can be used to represent the truth enables students to reflect critically on the information that societies are given or generate, and how this influences the allocation of resources or the choices that people make. This systematic analysis and critical reflection when problem solving may be inspiring springboards for CAS projects.

Students may also draw on their CAS experiences to enrich their involvement in mathematics both within and outside the classroom, and mathematics teachers can assist students in making links between their subjects and students' CAS experiences where appropriate. Purposeful discussion about real CAS experiences and projects will help students to make these links.

The challenge and enjoyment of CAS can often have a profound effect on mathematics students, who might choose, for example, to engage with CAS in the following ways:

- plan, write and implement a "mathematics scavenger hunt" where younger students tour the school answering interesting mathematics questions as part of their introduction to a new school
- as a CAS project students could plan and carry out a survey, create a database and analyse the results, and make suggestions to resolve a problem in the students' local area. This might be, for example, surveying the availability of fresh fruit and vegetables within a community, preparing an action plan with suggestions of how to increase availability or access, and presenting this to a local charity or community group
- taking an element of world culture that interests students and designing a miniature Earth (if the world were 100 people) to express the trend(s) numerically.

It is important to note that a CAS experience can be a single event or may be an extended series of events. However, CAS experiences must be distinct from, and may not be included or used in, the student's Diploma course requirements.

Additional suggestions on the links between DP subjects and CAS can be found in the *Creativity, activity, service teacher support material*.

Prior learning

It is expected that most students embarking on a DP mathematics course will have studied mathematics for at least 10 years. There will be a great variety of topics studied, and differing approaches to teaching and learning. Thus, students will have a wide variety of skills and knowledge when they start their mathematics course. Most will have some background in arithmetic, algebra, geometry, trigonometry, probability and statistics. Some will be familiar with an inquiry approach, and may have had an opportunity to complete an extended piece of work in mathematics.

At the beginning of the syllabus section there is a list of topics that are considered to be prior learning for the mathematics courses. It is recognized that this may contain topics that are unfamiliar to some students, but it is anticipated that there may be other topics in the syllabus itself which these students have already encountered. Teachers should be informed by their assessment of students' prior learning to help plan their teaching so that topics mentioned that are unfamiliar to their students can be incorporated.

Links to the Middle Years Programme

The MYP mathematics framework is designed to prepare students for the study of DP mathematics courses. As students progress from the MYP to the DP or the Career-related Programme (CP) they continue to develop their mathematics skills and knowledge, which will then allow them to go on to study a wide range of topics. Inquiry-based learning is central to mathematics courses in both MYP and the DP, providing students with opportunities to independently and collaboratively investigate, problem solve and communicate their mathematics with an increasing level of sophistication.

MYP mathematics courses are concept-driven, aimed at helping the learner to construct meaning through improved critical thinking and the transfer of knowledge. The MYP courses use a framework of key concepts with which the concepts in the DP mathematics courses are aligned. These concepts are broad, organizing, powerful ideas that have relevance within the subject but also transcend it, having relevance in other

subject groups. The fundamental concepts of MYP mathematics provide a very useful foundation for students following DP mathematics courses.

The aims of the MYP mathematics courses align very closely with those of the DP mathematics courses. The prior learning topics for the DP mathematics courses have been written in conjunction with the MYP mathematics guide.

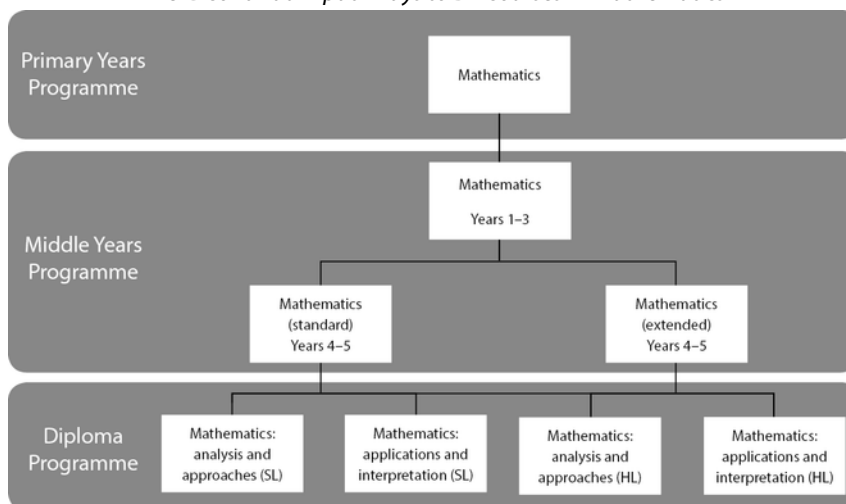
The MYP mathematics assessment objectives and criteria have been developed with both the internal and external assessment requirements of the DP in mind. MYP mathematics students are required to practise and develop their investigation skills, one of the MYP four assessment objectives, giving an important foundation for the internal assessment component of the DP mathematics courses. The MYP assessment objective of thinking critically also corresponds to the higher-order assessment objectives of communication and interpretation, and of reasoning that are expected from a DP mathematics student.

MYP and DP mathematics courses emphasized the use of technology as a powerful tool for learning, applying and communicating mathematics.

Where students in the MYP may select either standard or extended mathematics, at DP there are two mathematics subjects both available at SL and HL. MYP students enrolled in extended mathematics generally elect to take one of the HL mathematics courses in the DP. Students in MYP standard mathematics should seek the recommendation of their teacher when deciding which SL or HL course they are best suited to pursue.

Figure 3

The IB continuum pathways to DP courses in mathematics



Links to the Career-related Programme

In the IB Career-related Programme (CP) students study at least two DP subjects, a core consisting of four components and a career-related study, which is determined by the local context and aligned with student needs. The CP has been designed to add value to the student's career-related studies. This provides the context for the choice of DP courses. Courses can be chosen from any group of the DP. It is also possible to study more than one course from the same group (for example, visual arts and film).

Mathematics may be a beneficial choice for CP students considering careers in, for example, finance, planning, healthcare systems or coding, tourism industries, the technology industry, social informatics, or urban planning. Mathematics helps students to understand the value of systematic approaches, how to analyse complex real-world contexts, how to communicate this concisely and precisely and understand the implications of conclusions.

Mathematics encourages the development of strong written, verbal, and graphical communication skills; critical and complex thinking; and moral and ethical considerations influenced by mathematics that will assist students in preparing for the future global workplace. This in turn fosters the IB learner profile attributes that are transferable to the entire CP, providing relevance and support for the student's learning.

For the CP students, DP courses can be studied at SL or HL. Schools can explore opportunities to integrate CP students with DP students.

Approaches to the teaching and learning of mathematics: analysis and approaches

Conceptual understanding

Concepts are broad, powerful, organizing ideas, the significance of which goes beyond particular origins, subject matter or place in time. Concepts represent the vehicle for students' inquiry into issues and ideas of personal, local and global significance, providing the means by which they can explore the essence of mathematics.

Concepts play an important role in mathematics, helping students and teachers to think with increasing complexity as they organize and relate facts and topics. Students use conceptual understandings as they solve problems, analyse issues and evaluate decisions that can have an impact on themselves, their communities and the wider world.

In DP mathematics courses, conceptual understandings are key to promoting deep learning. The course identifies twelve fundamental concepts which relate with varying emphasis to each of the five topics. Teachers may identify and develop additional concepts to meet local circumstances and national or state curriculum requirements. Teachers can use these concepts to develop connections throughout the curriculum.

Each topic in this guide begins by stating the essential understandings of the topic and highlighting relevant concepts fundamental to the topic. This is followed by suggested conceptual understandings relevant to the content within the topic, although this list is not intended to be prescriptive or exhaustive.

The concepts

Concepts promote the development of a broad, balanced, conceptual and connected curriculum. They represent big ideas that are relevant and facilitate connections within topics, across topics and also to other subjects within the DP.

The twelve concepts identified below support conceptual understanding, can inform units of work and can help to organize teaching and learning. Explanations of each of these concepts in a mathematical context have also been provided.

Approximation	This concept refers to a quantity or a representation which is nearly but not exactly correct.
Change	This concept refers to a variation in size, amount or behaviour.
Equivalence	This concept refers to the state of being identically equal or interchangeable, applied to statements, quantities or expressions.
Generalization	This concept refers to a general statement made on the basis of specific examples.
Modelling	This concept refers to the way in which mathematics can be used to represent the real world.
Patterns	This concept refers to the underlying order, regularity or predictability of the elements of a mathematical system.
Quantity	This concept refers to an amount or number.
Relationships	This concept refers to the connection between quantities, properties or concepts; these connections may be expressed as models, rules or statements. Relationships provide opportunities for students to explore patterns in the world around them.

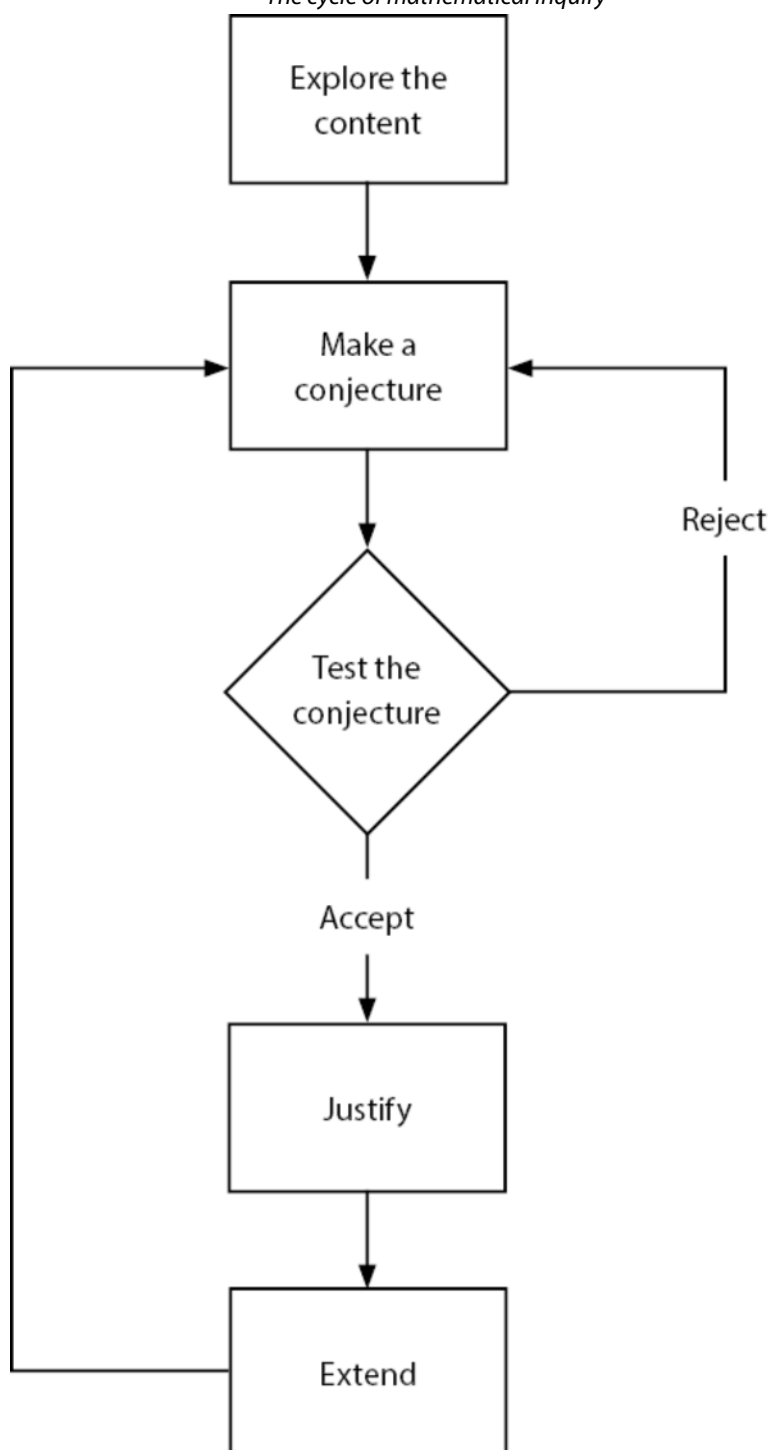
Representation	This concept refers to using words, formulae, diagrams, tables, charts, graphs and models to represent mathematical information.
Space	This concept refers to the frame of geometrical dimensions describing an entity.
Systems	This concept refers to groups of interrelated elements.
Validity	This concept refers to using well-founded, logical mathematics to come to a true and accurate conclusion or a reasonable interpretation of results.

Mathematical inquiry

Approaches to teaching and learning in the DP refer to deliberate strategies, skills and attitudes that permeate the teaching and learning environment. These approaches and tools are intrinsically linked to the IB learner profile, which encourages learning by experimentation, questioning and discovery.

In the IB classroom, students should regularly learn mathematics by being active participants in learning activities. Teachers should therefore provide students with regular opportunities to learn through mathematical inquiry, by making frequent use of strategies which stimulate students' critical thinking and problem-solving skills.

Figure 4

The cycle of mathematical inquiry

Mathematical modelling

Mathematical modelling is an important technique used in problem solving, to make sense of the real world. It is often used to help us better understand a situation, to check the effects of change or to inform

decision-making. Engaging students in the mathematical modelling process provides them with such opportunities. It is one of the most useful mathematical skills that students will need to be successful in many non-mathematical as well as mathematical courses and careers.

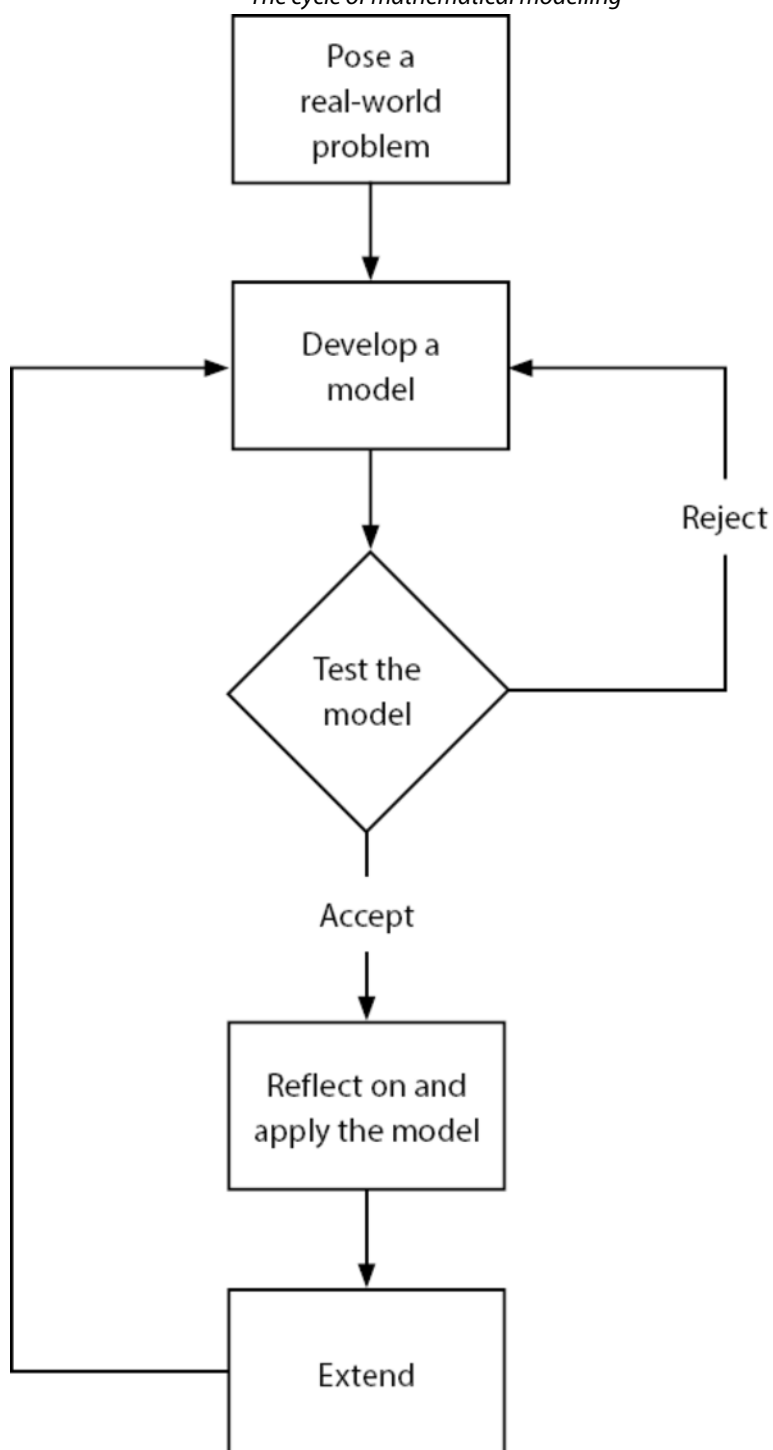
The mathematical modelling process begins with consideration of a situation that exists in the real world and is not usually artificially created. At this stage assumptions sometimes have to be made in order to simplify the situation to allow modelling to take place. There is often a fine balance needed between the simplicity and the accuracy of the model.

The first stage involves choosing or fitting a suitable mathematical representation to the context. This representation is tested to evaluate whether or not it returns expected results. The testing stage allows the results returned by the model to be reflected upon and adaptations to be made to the model if necessary. Once a satisfactory model is established it can be applied or used to explain a situation, check the effects of change or to inform decision-making.

The process of mathematical modelling requires critical reflection throughout the process.

More advice and guidance on the process of mathematical modelling is given in the teacher support material. The cycle of mathematical modelling is illustrated below.

Figure 5

The cycle of mathematical modelling

Proof

Proof in mathematics is an essential element in developing critical thinking. Engaging students in the process of proving a statement enables a deeper understanding of mathematical concepts. At standard

level students will be exposed to simple deductive proofs. In the additional higher level (AHL) content students will also focus on proof by contradiction and proof by induction as well as using counterexamples to show that a statement is not true.

The value of proving a statement is multifaceted as it helps students to develop the following skills:

- groupwork
- interpersonal skills
- reasoning
- research
- oral and written communication
- creative thinking
- organization.

Writing proofs enables students to appreciate proof techniques and mathematical thought processes. Students will learn the vocabulary and layout for proving mathematical statements. When faced with a mathematical statement, HL students will also be challenged to think about the best method to show that the statement is true. Proofs encourage students to reflect on mathematical rigour, efficiency and the elegance of showing that a statement is true.

Within the guide the term “informal” or “elementary” refers to approaches which do not require proof, can be justified with examples and do not make formal use of axioms.

Use of technology

The use of technology is an integral part of DP mathematics courses. Developing an appreciation of how developments in technology and mathematics have influenced each other is one of the aims of the courses and using technology accurately, appropriately and efficiently both to explore new ideas and to solve problems is one of the assessment objectives. Learning how to use different forms of technology is an important skill in mathematics and time has been allowed in each topic of the syllabus and through the “toolkit” in order to do this.

Technology is a powerful tool in mathematics and in recent years increased student and teacher access to this technology has supported and advanced the teaching and learning of mathematics. Discerning use of technology can make more mathematics accessible and motivating to a greater number of students.

Teachers can use technology to support and enhance student understanding in many ways including:

- to bring out teaching points
- to address misconceptions
- to aid visualization
- to enhance understanding of concepts that would otherwise be restricted by lengthy numerical calculations or algebraic manipulation
- to support students in making conjectures and checking generalizations
- to explicitly make the links between different mathematical representations or approaches.

Students can also use technology to engage with the learning process in many ways including the following:

- to develop and enhance their own personal conceptual understanding
- to search for patterns
- to test conjectures or generalizations
- to justify interpretations
- to collaborate on project based work
- to help organize and analyse data.

In the classroom teachers and students can use technology, working individually or collaboratively, to explore mathematical concepts. The key to successful learning of mathematics with technology is the fine

balance between the teacher and student use of technology, with carefully chosen use of technology to support the understanding and the communication of the mathematics itself.

Many topics within the DP mathematics courses lend themselves to the use of technology. Graphical calculators, dynamic graphing software, spreadsheets, simulations, apps, dynamic geometry software and interactive whiteboard software are just a few of the many kinds of technology available to support the teaching and learning of mathematics.

Within the guide the term “technology” is used for any form of calculator, hardware or software that may be available in the classroom. The terms “analysis” and “analytic approach” are generally used in the guide to indicate an algebraic approach that may not require the use of technology. It is important to note there will be restrictions on which technology may be used in examinations, which will be detailed in relevant documents.

As teachers tie together the unifying themes of **mathematical inquiry**, **mathematical modelling** and the **use of technology**, they should begin by providing substantial guidance and then gradually encourage students to become more independent as inquirers and thinkers. DP students should learn to become strong communicators through the language of mathematics. Teachers should create a learning environment in which students are comfortable as risk takers and inquiry, conceptual understanding, collaboration and use of technology feature prominently.

For further information on “Approaches to teaching and learning in the Diploma Programme”, please refer to the publication *The Diploma Programme: From principles into practice*. To support teachers, a variety of resources can be found on the programme resource centre, and details of workshops for professional development are available on the public website.

Format of the syllabus

The format of the syllabus section of the mathematics guides is the same for each subject and each level. This structure gives prominence and focus to the aspects of teaching and learning, including conceptual understandings, content and enrichment.

There are five topics and within these topics there are sub-topics. The five topics are:

- number and algebra
- functions
- geometry and trigonometry
- probability and statistics
- calculus

Each topic begins with a section on conceptual understandings. Details are given of which of the twelve key concepts could be used to relate to the topic. Essential understandings give details of the overall aims of the topic and then content-specific conceptual understandings give specific details of the aims and purpose of the topic and sub-topics.

Each topic begins with SL content which is common to both Mathematics: analysis and approaches and to Mathematics: applications and interpretation.

Each topic has SL content followed by AHL content. Teachers should ensure that they cover all SL content with SL students, and all SL and AHL content with HL students. The topics are structured so that an informal introduction in the common content may be formalized in the SL content and then extended further in the AHL content. For example the extension of the set of numbers for which something is defined could be integers in the common content, positive real numbers in the SL content and all real and complex numbers in the AHL content.

The content of all five topics at the appropriate level must be taught, however not necessarily in the order in which they appear in the guide. Teachers are expected to construct a course of study that addresses the needs of their students and includes, where necessary, the topics noted in prior learning. Guidance on structuring a course is given in the teacher support material.

Each topic has three sections:

Content: The column on the left contains details of the sub-topics to be covered.

Guidance, clarification and syllabus links: The column on the right contains more detailed information on specific sub-topics listed in the content column. This clarifies the content for the examination and highlights where sub-topics relate to other sub-topics within the syllabus.

Connections: Each topic also contains a short section that provides suggestions for further discussion, including real-life examples and ideas for further investigation.

These suggestions are only a guide and are not exhaustive. A downloadable version of these sections is also available, so that additional connections can be added to the ones already suggested by the IB. Potential areas for connections include:

- **Other contexts:** Real-life examples
- **Links to other subjects:** Suggested connections to other subjects within the DP. Note that these are correct for the current (2019) published versions of the guides
- **Aim:** Links to the aims of the course
- **International-mindedness:** Suggestions for discussions
- **TOK:** Suggestions for discussions
- **Links to TSM:** Links to the teacher support materials (TSM) in the “in practice” section of the website
- **Link to specimen paper:** Links to specific questions exemplifying how topics might be examined
- **Use of technology:** Suggestions as to how technology can be used in the classroom to enhance understanding
- **Links to websites:** Suggested websites for use in teaching or learning activities
- **Enrichment:** Suggestions for further discussions that may reinforce understanding.

Planning your course

The syllabus as provided in the subject guide is not intended to be a teaching order. Instead it provides detail of what must be covered by the end of the course. A school should develop a scheme of work that best works for its own students. For example, the scheme of work could be developed to match available resources, to take into account prior learning and experience, or in conjunction with other local requirements.

HL teachers may choose to teach the SL content and AHL content at the same time or teach them in a spiral fashion, by teaching the SL content first and revisiting these topics through the delivery of the AHL topics after this.

However the course is planned, adequate time must be provided for examination revision. Time must also be given to students to reflect on their learning experience and their growth as learners.

Time allocation

The recommended teaching time for HL courses is 240 hours and for SL courses is 150 hours. For mathematics courses at both SL and HL, it is expected that 30 hours will be spent on developing inquiry, modelling and investigation skills. This includes up to 15 hours for work on the internal assessment which is called the exploration. The time allocations given in this guide are approximate, and are intended to suggest how the remaining 210 hours for HL and 120 hours for SL allowed for the teaching of the syllabus might be allocated. The exact time spent on each topic depends on a number of factors, including the background knowledge and level of preparedness of each student. Teachers should therefore adjust these timings to correspond to the needs of their students.

The Toolkit

Time has been allocated within the teaching hours for students to undertake the types of activities that mathematicians in the real world undertake and to allow students time to develop the skill of thinking like a

mathematician—in other words providing students with a mathematical toolkit which will allow them to approach any type of mathematical problem. Underpinning this are the six pedagogical approaches to teaching and the five approaches to learning which support all IB programmes. This time gives students opportunities in the classroom for undertaking an inquiry-based approach and focusing on conceptual understanding of the content, developing their awareness of mathematics in local and global contexts, gives them opportunities for teamwork and collaboration as well as time to reflect upon their own learning of mathematics.

Students should be encouraged to actively identify skills that they might add to their personal mathematics toolkit. Teachers are encouraged to make explicit where these skills might transfer across areas of mathematics content and allow students to reflect upon where these skills transfer to other subjects the student is studying.

The teacher support material (TSM) contains a section referred to as the “toolkit”. This section contains ideas and resources that teachers can use with their students to encourage the development of mathematical thinking skills. These resources have been developed by teachers for use in their own classrooms and are not exhaustive.

Formulae and the formula booklet

Formulae are only included in this guide document where there may be some ambiguity. All formulae required for the course are in the mathematics formula booklet.

From the beginning of the course it is recommended that teachers ensure students are familiar with the contents of the formula booklet by either giving students a printed copy or making an electronic copy available to them.

Each student is required to have access to a clean copy of the formula booklet during the examination. For each examination, it is the responsibility of the school to download a copy of the formula booklet from IBIS or the programme resource centre, check that there are no printing errors, and ensure that there are sufficient copies available for all students.

Command terms and notation list

Teachers and students need to be familiar with the IB notation list and the command terms, as these will be used without explanation in the examination papers. The “Glossary of command terms” and “Notation list” appear as appendices in this guide.

Aims

The aims of all DP mathematics courses are to enable students to:

1. develop a curiosity and enjoyment of mathematics, and appreciate its elegance and power
2. develop an understanding of the concepts, principles and nature of mathematics
3. communicate mathematics clearly, concisely and confidently in a variety of contexts
4. develop logical and creative thinking, and patience and persistence in problem solving to instil confidence in using mathematics
5. employ and refine their powers of abstraction and generalization
6. take action to apply and transfer skills to alternative situations, to other areas of knowledge and to future developments in their local and global communities
7. appreciate how developments in technology and mathematics influence each other
8. appreciate the moral, social and ethical questions arising from the work of mathematicians and the applications of mathematics
9. appreciate the universality of mathematics and its multicultural, international and historical perspectives
10. appreciate the contribution of mathematics to other disciplines, and as a particular “area of knowledge” in the TOK course
11. develop the ability to reflect critically upon their own work and the work of others
12. independently and collaboratively extend their understanding of mathematics.

Assessment objectives

Problem solving is central to learning mathematics and involves the acquisition of mathematical skills and concepts in a wide range of situations, including non-routine, open-ended and real-world problems. Having followed a DP mathematics course, students will be expected to demonstrate the following:

1. **Knowledge and understanding:** Recall, select and use their knowledge of mathematical facts, concepts and techniques in a variety of familiar and unfamiliar contexts.
2. **Problem solving:** Recall, select and use their knowledge of mathematical skills, results and models in both abstract and real-world contexts to solve problems.
3. **Communication and interpretation:** Transform common realistic contexts into mathematics; comment on the context; sketch or draw mathematical diagrams, graphs or constructions both on paper and using technology; record methods, solutions and conclusions using standardized notation; use appropriate notation and terminology.
4. **Technology:** Use technology accurately, appropriately and efficiently both to explore new ideas and to solve problems.
5. **Reasoning:** Construct mathematical arguments through use of precise statements, logical deduction and inference and by the manipulation of mathematical expressions.
6. **Inquiry approaches:** Investigate unfamiliar situations, both abstract and from the real world, involving organizing and analyzing information, making conjectures, drawing conclusions, and testing their validity.

Assessment objectives in practice

Assessment objectives	Paper 1 %	Paper 2 %	Paper 3 % HL only	Exploration %
Knowledge and understanding	20-30	15-25	10-20	5-15
Problem solving	20-30	15-25	20-30	5-20
Communication and interpretation	20-30	15-25	15-25	15-25
Technology	0	25-35	10-30	10-20
Reasoning	5-15	5-10	10-20	5-25
Inquiry approaches	10-20	5-10	15-30	25-35

Syllabus outline

Syllabus component	Suggested teaching hours	
	SL	HL
Topic 1—Number and algebra	19	39
Topic 2—Functions	21	32
Topic 3— Geometry and trigonometry	25	51
Topic 4—Statistics and probability	27	33
Topic 5 —Calculus	28	55
The toolkit and the mathematical exploration Investigative, problem-solving and modelling skills development leading to an individual exploration. The exploration is a piece of written work that involves investigating an area of mathematics.	30	30
Total teaching hours	150	240

All topics are compulsory. Students must study all the sub-topics in each of the topics in the syllabus as listed in this guide. Students are also required to be familiar with the topics listed as prior learning.

Prior learning topics

Prior to starting a DP mathematics course students have extensive previous mathematical experiences, but these will vary. It is expected that mathematics students will be familiar with the following topics before they take the examinations because questions assume knowledge of them. Teachers must therefore ensure that any topics listed here that are unknown to their students at the start of the course are included at an early stage. Teachers should also take into account the existing mathematical knowledge of their students to design an appropriate course of study for mathematics. This table lists the knowledge, together with the syllabus content, that is essential for successful completion of the mathematics course.

Number and algebra

- Number systems: natural numbers $\mathbb{N}$; integers, $\mathbb{Z}$; rationals, $\mathbb{Q}$, and irrationals; real numbers, $\mathbb{R}$
- SI (Système International) units for mass, time, length, area and volume and their derived units, eg. speed
- Rounding, decimal approximations and significant figures, including appreciation of errors
- Definition and elementary treatment of absolute value (modulus), $|a|$
- Use of addition, subtraction, multiplication and division using integers, decimals and fractions, including order of operations
- Prime numbers, factors (divisors) and multiples
- Greatest common factor (divisor) and least common multiples (HL only)
- Simple applications of ratio, percentage and proportion
- Manipulation of algebraic expressions, including factorization and expansion
- Rearranging formulae
- Calculating the numerical value of expressions by substitution
- Evaluating exponential expressions with simple positive exponents
- Evaluating exponential expressions with rational exponents (HL only)
- Use of inequalities, $<$, $\leq$, $>$, $\geq$, intervals on the real number line
- Simplification of simple expressions involving roots (surds or radicals)
- Rationalising the denominator (HL only)
- Expression of numbers in the form $a \times 10^k$, $1 \leq a < 10$, $k \in \mathbb{Z}$
- Familiarity with commonly accepted world currencies
- Solution of linear equations and inequalities
- Solution of quadratic equations and inequalities with rational coefficients (HL only)
- Solving systems of linear equations in two variables
- Concept and basic notation of sets. Operations on sets: union and intersection
- Addition and subtraction of algebraic fractions (HL only).

Functions

- Graphing linear and quadratic functions using technology
- Mappings of the elements of one set to another. Illustration by means of sets of ordered pairs, tables, diagrams and graphs.

Geometry and trigonometry

- Pythagoras' theorem and its converse
- Mid-point of a line segment and the distance between two points in the Cartesian plane
- Geometric concepts: point, line, plane, angle
- Angle measurement in degrees, compass directions
- The triangle sum theorem
- Right-angle trigonometry, including simple applications for solving triangles
- Three-figure bearings
- Simple geometric transformations: translation, reflection, rotation, enlargement
- The circle, its centre and radius, area and circumference. The terms diameter, arc, sector, chord, tangent and segment
- Perimeter and area of plane figures. Properties of triangles and quadrilaterals, including parallelograms, rhombuses, rectangles, squares, kites and trapezoids; compound shapes
- Familiarity with three-dimensional shapes (prisms, pyramids, spheres, cylinders and cones)
- Volumes and surface areas of cuboids, prisms, cylinders, and compound three-dimensional shapes

Statistics and probability

- The collection of data and its representation in bar charts, pie charts, pictograms, and line graphs
- Obtaining simple statistics from discrete data, including mean, median, mode, range
- Calculating probabilities of simple events
- Venn diagrams for sorting data
- Tree diagrams

Calculus

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

Syllabus content

Topic 1: Number and algebra

Concepts

Essential understandings:

Number and algebra allow us to represent patterns, show equivalencies and make generalizations which enable us to model real-world situations. Algebra is an abstraction of numerical concepts and employs variables which allow us to solve mathematical problems.

Suggested concepts embedded in this topic:

Generalization, representation, modelling, equivalence, patterns, quantity

AHL: Validity, systems.

Content-specific conceptual understandings:

- Modelling real-life situations with the structure of arithmetic and geometric sequences and series allows for prediction, analysis and interpretation.
- Different representations of numbers enable equivalent quantities to be compared and used in calculations with ease to an appropriate degree of accuracy.
- Numbers and formulae can appear in different, but equivalent, forms, or representations, which can help us to establish identities.
- Formulae are a generalization made on the basis of specific examples, which can then be extended to new examples.
- Logarithm laws provide the means to find inverses of exponential functions which model real-life situations.
- Patterns in numbers inform the development of algebraic tools that can be applied to find unknowns.
- The binomial theorem is a generalization which provides an efficient method for expanding binomial expressions.

AHL

- Proof serves to validate mathematical formulae and the equivalence of identities.
- Representing partial fractions and complex numbers in different forms allows us to easily carry out seemingly difficult calculations.
- The solution for systems of equations can be carried out by a variety of equivalent algebraic and graphical methods.

SL content

Recommended teaching hours: 19

The aim of the SL content of the number and algebra topic is to introduce students to numerical concepts and techniques which, combined with an introduction to arithmetic and geometric sequences and series, can be used for financial and other applications. Students will also be introduced to the formal concept of proof.

Sections SL1.1 to SL1.5 are content common to Mathematics: analysis and approaches and Mathematics: applications and interpretation.

SL 1.1

Content	Guidance, clarification and syllabus links
Operations with numbers in the form $a \times 10^k$ where $1 \leq a < 10$ and k is an integer.	Calculator or computer notation is not acceptable. For example, 5.2E30 is not acceptable and should be written as 5.2×10^{30} .

Connections

Other contexts: Very large and very small numbers, for example astronomical distances, sub-atomic particles in physics, global financial figures

Links to other subjects: Chemistry (Avogadro's number); physics (order of magnitude); biology (microscopic measurements); sciences group subjects (uncertainty and precision of measurement)

International-mindedness: The history of number from Sumerians and its development to the present Arabic system

TOK: Do the names that we give things impact how we understand them? For instance, what is the impact of the fact that some large numbers are named, such as the googol and the googolplex, while others are represented in this form?

[Download connections template](#)

SL 1.2

Content	Guidance, clarification and syllabus links
Arithmetic sequences and series. Use of the formulae for the n^{th} term and the sum of the first n terms of the sequence. Use of sigma notation for sums of arithmetic sequences.	Spreadsheets, GDCs and graphing software may be used to generate and display sequences in several ways. If technology is used in examinations, students will be expected to identify the first term and the common difference.
Applications.	Examples include simple interest over a number of years.
Analysis, interpretation and prediction where a model is not perfectly arithmetic in real life.	Students will need to approximate common differences.

Connections

International-mindedness: The chess legend (Sissa ibn Dahir); Aryabhata is sometimes considered the "father of algebra"—compare with alKhawarizmi; the use of several alphabets in mathematical notation (for example the use of capital sigma for the sum).

TOK: Is all knowledge concerned with identification and use of patterns? Consider Fibonacci numbers and connections with the golden ratio.

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SL 1.3

Content	Guidance, clarification and syllabus links
Geometric sequences and series. Use of the formulae for the n^{th} term and the sum of the first n terms of the sequence.	Spreadsheets, GDCs and graphing software may be used to generate and display sequences in several ways.

Content	Guidance, clarification and syllabus links
Use of sigma notation for the sums of geometric sequences.	If technology is used in examinations, students will be expected to identify the first term and the ratio. Link to: models/functions in topic 2 and regression in topic 4.
Applications.	Examples include the spread of disease, salary increase and decrease and population growth.

Connections

Links to other subjects: Radioactive decay, nuclear physics, charging and discharging capacitors (physics).

TOK: How do mathematicians reconcile the fact that some conclusions seem to conflict with our intuitions? Consider for instance that a finite area can be bounded by an infinite perimeter.

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SL 1.4

Content	Guidance, clarification and syllabus links
Financial applications of geometric sequences and series: • compound interest • annual depreciation.	Examination questions may require the use of technology, including built-in financial packages. The concept of simple interest may be used as an introduction to compound interest. Calculate the real value of an investment with an interest rate and an inflation rate. In examinations, questions that ask students to derive the formula will not be set. Compound interest can be calculated yearly, half-yearly, quarterly or monthly. Link to: exponential models/functions in topic 2.

Connections

Other contexts: Loans.

Links to other subjects: Loans and repayments (economics and business management).

Aim 8: Ethical perceptions of borrowing and lending money.

International-mindedness: Do all societies view investment and interest in the same way?

TOK: How have technological advances affected the nature and practice of mathematics? Consider the use of financial packages for instance.

Enrichment: The concept of e can be introduced through continuous compounding, $(1 + \frac{1}{n})^n \rightarrow e$, as $n \rightarrow \infty$, however this will not be examined.

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SL 1.5

Content	Guidance, clarification and syllabus links
Laws of exponents with integer exponents.	Examples:

Content	Guidance, clarification and syllabus links
	$5^3 \times 5^{-6} = 5^{-3}, 6^4 \div 6^3 = 6, (2^3)^{-4} = 2^{-12},$ $(2x)^4 = 16x^4, 2x^{-3} = \frac{2}{x^3}.$
Introduction to logarithms with base 10 and e. Numerical evaluation of logarithms using technology.	Awareness that $a^x = b$ is equivalent to $\log_a b = x$, that $b > 0$, and $\log_e x = \ln x$.

Connections

Other contexts: Richter scale and decibel scale.

Links to other subjects: Calculation of pH and buffer solutions (chemistry)

TOK: Is mathematics invented or discovered? For instance, consider the number e or logarithms—did they already exist before man defined them? (This topic is an opportunity for teachers to generate reflection on “the nature of mathematics”).

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SL 1.6

Content	Guidance, clarification and syllabus links
Simple deductive proof, numerical and algebraic; how to lay out a left-hand side to right-hand side (LHS to RHS) proof. The symbols and notation for equality and identity.	Example: Show that $\frac{1}{4} + \frac{1}{12} = \frac{1}{3}$. Show that the algebraic generalisation of this is $\frac{1}{m+1} + \frac{1}{m^2+m} \equiv \frac{1}{m}$ LHS to RHS proofs require students to begin with the left-hand side expression and transform this using known algebraic steps into the expression on the right-hand side (or vice versa). Example: Show that $(x-3)^2 + 5 \equiv x^2 - 6x + 14$. Students will be expected to show how they can check a result including a check of their own results.

Connections

TOK: Is mathematical reasoning different from scientific reasoning, or reasoning in other Areas of Knowledge?

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SL 1.7

Content	Guidance, clarification and syllabus links
Laws of exponents with rational exponents.	$\frac{1}{a^m} = \sqrt[m]{a}$, if m is even this refers to the positive root. For example: $16^{\frac{3}{4}} = 8$.
Laws of logarithms. $\log_a xy = \log_a x + \log_a y$ $\log_a \frac{x}{y} = \log_a x - \log_a y$	$y = a^x \Leftrightarrow x = \log_a y; \log_a a = 1, \log_a 1 = 0,$ $a, y \in \mathbb{N}, x \in \mathbb{Z}$ Link to: introduction to logarithms (SL1.5)

Content	Guidance, clarification and syllabus links
$\log_a x^m = m \log_a x$ for $a, x, y > 0$	Examples: $\frac{3}{4} = \log_{16} 8, \log_{32} 2 = 5 \log 2$ $\log 24 = \log 8 + \log 3$ $\log_3 \frac{10}{4} = \log_3 10 - \log_3 4$ $\log_4 3^5 = 5 \log_4 3$ Link to: logarithmic and exponential graphs (SL2.9)
Change of base of a logarithm. $\log_a x = \frac{\log_b x}{\log_b a}, \text{ for } a, b, x > 0$	$\log_4 7 = \frac{\ln 7}{\ln 4}$ Examples: $\log_{25} 125 = \frac{\log_5 125}{\log_5 25} \quad (= \frac{3}{2})$
Solving exponential equations, including using logarithms.	Examples: $(\frac{1}{3})^x = 9^{x+1}, 2^{x-1} = 10.$ Link to: using logarithmic and exponential graphs (SL2.9).

Connections

Links to other subjects: pH, buffer calculations and finding activation energy from experimental data (chemistry).

TOK: How have seminal advances, such as the development of logarithms, changed the way in which mathematicians understand the world and the nature of mathematics?

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SL 1.8

Content	Guidance, clarification and syllabus links
Sum of infinite convergent geometric sequences.	Use of $ r < 1$ and modulus notation. Link to: geometric sequences and series (SL1.3).

Connections

TOK: Is it possible to know about things of which we can have no experience, such as infinity?

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SL 1.9

Content	Guidance, clarification and syllabus links
The binomial theorem: expansion of $(a + b)^n, n \in \mathbb{N}$.	Counting principles may be used in the development of the theorem.
Use of Pascal's triangle and nC_r .	nC_r should be found using both the formula and technology. Example: Find r when ${}^6C_r = 20$, using a table of values generated with technology.

Connections

Aim 8: Ethics in mathematics–Pascal’s triangle. Attributing the origin of a mathematical discovery to the wrong mathematician.

International-mindedness: The properties of “Pascal’s triangle” have been known in a number of different cultures long before Pascal. (for example the Chinese mathematician Yang Hui).

TOK: How have notable individuals shaped the development of mathematics as an area of knowledge? Consider Pascal and “his” triangle.

[Download connections template](#)

AHL content

Recommended teaching hours: 20

The aim of the AHL content in the number and algebra topic is to extend and build upon the aims, concepts and skills from the SL content. It introduces students to some important techniques for expansion, simplification and solution of equations. Complex numbers are introduced and students will extend their knowledge of formal proof to proof by mathematical induction, proof by contradiction and proof by counterexample.

AHL 1.10

Content	Guidance, clarification and syllabus links
Counting principles, including permutations and combinations.	Not required: Permutations where some objects are identical. Circular arrangements.
Extension of the binomial theorem to fractional and negative indices, ie $(a + b)^n$, $n \in \mathbb{Q}$.	$(a + b)^n = \left(a\left(1 + \frac{b}{a}\right)\right)^n = a^n\left(1 + \frac{b}{a}\right)^n, n \in \mathbb{Q}$ Link to: power series expansions (AHL5.19) Not required: Proof of binomial theorem.

Connections

Other contexts: Finding approximations to $\sqrt{2}$

Aim 8: How many different tickets are possible in a lottery? What does this tell us about the ethics of selling lottery tickets to those who do not understand the implications of these large numbers?

International-mindedness: The properties of “Pascal’s triangle” have been known in a number of different cultures long before Pascal (for example the Chinese mathematician Yang Hui).

TOK: What counts as understanding in mathematics? Is it more than just getting the right answer?

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AHL 1.11

Content	Guidance, clarification and syllabus links
Partial fractions.	Maximum of two distinct linear terms in the denominator, with degree of numerator less than the degree of the denominator. Example: $\frac{2x + 1}{x^2 + x - 2} \equiv \frac{1}{(x - 1)} + \frac{1}{(x + 2)}$. Link to: use of partial fractions to rearrange the integrand (AHL5.15).

Connections
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AHL 1.12

Content	Guidance, clarification and syllabus links
Complex numbers: the number i , where $i^2 = -1$. Cartesian form $z = a + bi$; the terms real part, imaginary part, conjugate, modulus and argument.	
The complex plane.	The complex plane is also known as the Argand diagram. Link to: vectors (AHL3.12).

Connections

Other contexts: Concepts in electrical engineering—impedance as a combination of resistance and reactance, also apparent power as a combination of real and reactive powers. These combinations take the form $a + bi$.

TOK: How does language shape knowledge? For example, do the words “imaginary” and “complex” make the concepts more difficult than if they had different names?

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AHL 1.13

Content	Guidance, clarification and syllabus links
Modulus–argument (polar) form: $z = r(\cos\theta + i\sin\theta) = r\operatorname{cis}\theta$ Euler form: $z = re^{i\theta}$ Sums, products and quotients in Cartesian, polar or Euler forms and their geometric interpretation.	The ability to convert between Cartesian, modulus–argument (polar) and Euler form is expected.

Connections

Other contexts: Concepts in electrical engineering—phase angle/shift, power factor and apparent power as a complex quantity in polar form.

TOK: Why might it be said that $e^{i\pi} + 1 = 0$ is beautiful? What is the place of beauty and elegance in mathematics? What about the place of creativity?

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AHL 1.14

Content	Guidance, clarification and syllabus links
Complex conjugate roots of quadratic and polynomial equations with real coefficients.	Complex roots occur in conjugate pairs.
De Moivre’s theorem and its extension to rational exponents.	Includes proof by induction for the case where $n \in \mathbb{Z}^+$; awareness that it is true for $n \in \mathbb{R}$.

Content	Guidance, clarification and syllabus links
Powers and roots of complex numbers.	Link to: sum and product of roots of polynomial equations (AHL 2.12), compound angle identities (AHL 3.10).

Connections

TOK: Could we ever reach a point where everything important in a mathematical sense is known? Reflect on the creation of complex numbers before their applications were known.

Enrichment: Can De Moivre's theorem be extended to all n ?

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AHL 1.15

Content	Guidance, clarification and syllabus links
Proof by mathematical induction.	Proof should be incorporated throughout the course where appropriate. Mathematical induction links specifically to a wide variety of topics, for example complex numbers, differentiation, sums of sequences and divisibility.
Proof by contradiction.	Examples: Irrationality of $\sqrt{3}$; irrationality of the cube root of 5; Euclid's proof of an infinite number of prime numbers; if a is a rational number and b is an irrational number, then $a + b$ is an irrational number.
Use of a counterexample to show that a statement is not always true.	Example: Consider the set P of numbers of the form $n^2 + 41n + 41$, $n \in \mathbb{N}$, show that not all elements of P are prime. Example: Show that the following statement is not always true: there are no positive integer solutions to the equation $x^2 + y^2 = 10$. It is not sufficient to state the counterexample alone. Students must explain why their example is a counterexample.

Connections

Other contexts: The Four-colour theorem

International-mindedness: How did the Pythagoreans find out that $\sqrt{2}$ is irrational?

TOK: What is the role of the mathematical community in determining the validity of a mathematical proof? Do proofs provide us with completely certain knowledge? What is the difference between the inductive method in science and proof by induction in mathematics?

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AHL 1.16

Content	Guidance, clarification and syllabus links
Solutions of systems of linear equations (a maximum of three equations in three unknowns),	These systems should be solved using both algebraic and technological methods, for example row reduction or matrices.

Content	Guidance, clarification and syllabus links
including cases where there is a unique solution, an infinite number of solutions or no solution.	Systems which have no solution(s) are inconsistent. Finding a general solution for a system with an infinite number of solutions. Link to: intersection of lines and planes (AHL 3.18).

Connections

TOK: Mathematics, Sense, Perception and Reason: If we can find solutions in higher dimensions can we reason that these spaces exist beyond our sense perception?

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Topic 2: Functions

Concepts

Essential understandings

Models are depictions of real-life events using expressions, equations or graphs while a function is defined as a relation or expression involving one or more variables. Creating different representations of functions to model the relationships between variables, visually and symbolically as graphs, equations and tables represents different ways to communicate mathematical ideas.

Suggested concepts embedded in this topic:

Representation, relationships, space, quantity, equivalence.

AHL: Systems, patterns.

Content-specific conceptual understandings:

- Different representations of functions, symbolically and visually as graphs, equations and tables provide different ways to communicate mathematical relationships.
- The parameters in a function or equation correspond to geometrical features of a graph and can represent physical quantities in spatial dimensions.
- Moving between different forms to represent functions allows for deeper understanding and provides different approaches to problem solving.
- Our spatial frame of reference affects the visible part of a function and by changing this “window” can show more or less of the function to best suit our needs.
- Equivalent representations of quadratic functions can reveal different characteristics of the same relationship.
- Functions represent mappings that assign to each value of the independent variable (input) one and only one dependent variable (output).

AHL

- Extending results from a specific case to a general form can allow us to apply them to a larger system.
- Patterns can be identified in behaviours which can give us insight into appropriate strategies to model or solve them.
- The intersection of a system of equations may be represented graphically and algebraically and represents the solution that satisfies the equations.

SL content

Recommended teaching hours: 21

The aim of the SL content in the functions topic is to introduce students to the important unifying theme of a function in mathematics and to apply functional methods to a variety of mathematical situations.

Throughout this topic students should be given the opportunity to use technology, such as graphing packages and graphing calculators to develop and apply their knowledge of functions, rather than using elaborate analytic techniques.

On examination papers:

- questions may be set requiring the graphing of functions that do not explicitly appear on the syllabus
- the domain will be the largest possible domain for which a function is defined unless otherwise stated; this will usually be the real numbers

Sections SL2.1 to SL2.4 are content common to both Mathematics: analysis and approaches and Mathematics: applications and interpretation.

SL 2.1

Content	Guidance, clarification and syllabus links
Different forms of the equation of a straight line. Gradient; intercepts. Lines with gradients m_1 and m_2 Parallel lines $m_1 = m_2$. Perpendicular lines $m_1 \times m_2 = -1$.	$y = mx + c$ (gradient-intercept form). $ax + by + d = 0$ (general form). $y - y_1 = m(x - x_1)$ (point-gradient form). Calculate gradients of inclines such as mountain roads, bridges, etc.

Connections

Other contexts: Gradients of mountain roads, gradients of access ramps.

Links to other subjects: Exchange rates and price and income elasticity, demand and supply curves (economics); graphical analysis in experimental work (sciences group subjects).

TOK: Descartes showed that geometric problems could be solved algebraically and vice versa. What does this tell us about mathematical representation and mathematical knowledge?

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SL 2.2

Content	Guidance, clarification and syllabus links
Concept of a function, domain, range and graph. Function notation, for example $f(x)$, $v(t)$, $C(n)$. The concept of a function as a mathematical model.	Example: $f(x) = \sqrt{2 - x}$, the domain is $x \leq 2$, range is $f(x) \geq 0$. A graph is helpful in visualizing the range.
Informal concept that an inverse function reverses or undoes the effect of a function. Inverse function as a reflection in the line $y = x$, and the notation $f^{-1}(x)$.	Example: Solving $f(x) = 10$ is equivalent to finding $f^{-1}(10)$. Students should be aware that inverse functions exist for one to one functions; the domain of $f^{-1}(x)$ is equal to the range of $f(x)$.

Connections

Other contexts: Temperature and currency conversions.

Links to other subjects: Currency conversions and cost functions (economics and business management); projectile motion (physics).

Aim 8: What is the relationship between real-world problems and mathematical models?

International-mindedness: The development of functions by Rene Descartes (France), Gottfried Wilhelm Leibnitz (Germany) and Leonhard Euler (Switzerland); the notation for functions was developed by a number of different mathematicians in the 17th and 18th centuries—how did the notation we use today become internationally accepted?

TOK: Do you think mathematics or logic should be classified as a language?

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SL 2.3

Content	Guidance, clarification and syllabus links
The graph of a function; its equation $y = f(x)$.	Students should be aware of the difference between the command terms “draw” and “sketch”.
Creating a sketch from information given or a context, including transferring a graph from screen to paper.	All axes and key features should be labelled. This may include functions not specifically mentioned in topic 2.
Using technology to graph functions including their sums and differences.	

Connections

Links to other subjects: Sketching and interpreting graphs (sciences group subjects, geography, economics).

TOK: Does studying the graph of a function contain the same level of mathematical rigour as studying the function algebraically? What are the advantages and disadvantages of having different forms and symbolic language in mathematics?

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SL 2.4

Content	Guidance, clarification and syllabus links
Determine key features of graphs.	Maximum and minimum values; intercepts; symmetry; vertex; zeros of functions or roots of equations; vertical and horizontal asymptotes using graphing technology.
Finding the point of intersection of two curves or lines using technology.	

Connections

Links to other subjects: Identification and interpretation of key features of graphs (sciences group subjects, geography, economics); production possibilities curve model, market equilibrium (economics).

International-mindedness: Bourbaki group analytical approach versus the Mandelbrot visual approach.

Use of technology: Graphing technology with sliders to determine the effects of altering parameters and variables.

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SL 2.5

Content	Guidance, clarification and syllabus links
Composite functions.	$(f \circ g)(x) = f(g(x))$
Identity function. Finding the inverse function $f^{-1}(x)$.	$(f \circ f^{-1})(x) = (f^{-1} \circ f)(x) = x$ The existence of an inverse for one-to-one functions. Link to: concept of inverse function as a reflection in the line $y = x$ (SL 2.2).

Connections

TOK: Do you think mathematics or logic should be classified as a language?

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SL 2.6

Content	Guidance, clarification and syllabus links
The quadratic function $f(x) = ax^2 + bx + c$: its graph, y -intercept $(0, c)$. Axis of symmetry. The form $f(x) = a(x - p)(x - q)$, x -intercepts $(p, 0)$ and $(q, 0)$. The form $f(x) = a(x - h)^2 + k$, vertex (h, k) .	A quadratic graph is also called a parabola. Link to: transformations (SL 2.11). Candidates are expected to be able to change from one form to another.

Connections

Links to other subjects: Kinematics, projectile motion and simple harmonic motion (physics).

TOK: Are there fundamental differences between mathematics and other areas of knowledge? If so, are these differences more than just methodological differences?

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SL 2.7

Content	Guidance, clarification and syllabus links
Solution of quadratic equations and inequalities. The quadratic formula.	Using factorization, completing the square (vertex form), and the quadratic formula. Solutions may be referred to as roots or zeros.
The discriminant $\Delta = b^2 - 4ac$ and the nature of the roots, that is, two distinct real roots, two equal real roots, no real roots.	Example: For the equation $3kx^2 + 2x + k = 0$, find the possible values of k , which will give two distinct real roots, two equal real roots or no real roots.

Connections

Links to other subjects: Projectile motion and energy changes in simple harmonic motion (physics); equilibrium equations (chemistry).

International-mindedness: The Babylonian method of multiplication: $ab = \frac{(a+b)^2 - a^2 - b^2}{2}$. Sulba Sutras in ancient India and the Bakhshali Manuscript contained an algebraic formula for solving quadratic equations.

TOK: What are the key concepts that provide the building blocks for mathematical knowledge?

Use of technology: Dynamic graphing software with a slider.

Enrichment: Deriving the quadratic formula by completing the square.

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SL 2.8

Content	Guidance, clarification and syllabus links
The reciprocal function $f(x) = \frac{1}{x}$, $x \neq 0$: its graph and self-inverse nature.	
Rational functions of the form $f(x) = \frac{ax+b}{cx+d}$ and their graphs. Equations of vertical and horizontal asymptotes.	Sketches should include all horizontal and vertical asymptotes and any intercepts with the axes. Link to: transformations (SL2.11). Vertical asymptote: $x = -\frac{d}{c}$; Horizontal asymptote: $y = \frac{a}{c}$.

Connections

International-mindedness: The development of functions, Rene Descartes (France), Gottfried Wilhelm Leibniz (Germany) and Leonhard Euler (Switzerland).

TOK: What are the implications of accepting that mathematical knowledge changes over time?

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SL 2.9

Content	Guidance, clarification and syllabus links
Exponential functions and their graphs: $f(x) = a^x$, $a > 0$, $f(x) = e^x$ Logarithmic functions and their graphs: $f(x) = \log_a x$, $x > 0$, $f(x) = \ln x$, $x > 0$.	Link to: financial applications of geometric sequences and series (SL 1.4). Relationships between these functions: $a^x = e^{x \ln a}$; $\log_a a^x = x$, a , $x > 0$, $a \neq 1$ Exponential and logarithmic functions as inverses of each other.

Connections

Links to other subjects: Radioactive decay, charging and discharging capacitors (physics); first order reactions and activation energy (chemistry); growth curves (biology).

Aim 8: The phrase “exponential growth” is used popularly to describe a number of phenomena. Is this a misleading use of a mathematical term?

TOK: What role do “models” play in mathematics? Do they play a different role in mathematics compared to their role in other areas of knowledge?

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SL 2.10

Content	Guidance, clarification and syllabus links
Solving equations, both graphically and analytically.	Example: $e^{2x} - 5e^x + 4 = 0$. Link to: function graphing skills (SL 2.3).
Use of technology to solve a variety of equations, including those where there is no appropriate analytic approach.	Examples: $e^x = \sin x$ $x^4 + 5x - 6 = 0$.
Applications of graphing skills and solving equations that relate to real-life situations.	Link to: exponential growth (SL 2.9)

Connections

Other contexts: Radioactive decay and population growth and decay, compound interest, projectile motion, braking distances.

Links to other subjects: Radioactive decay (physics); modelling (sciences group subjects); production possibilities curve model (economics).

TOK: What assumptions do mathematicians make when they apply mathematics to real-life situations?

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SL 2.11

Content	Guidance, clarification and syllabus links
Transformations of graphs. Translations: $y = f(x) + b$; $y = f(x - a)$. Reflections (in both axes): $y = -f(x)$; $y = f(-x)$. Vertical stretch with scale factor p : $y = pf(x)$. Horizontal stretch with scale factor $\frac{1}{q}$: $y = f(qx)$.	Students should be aware of the relevance of the order in which transformations are performed. Dynamic graphing packages could be used to investigate these transformations.
Composite transformations.	Example: Using $y = x^2$ to sketch $y = 3x^2 + 2$ Link to: composite functions (SL2.5). Not required at SL: transformations of the form $f(ax + b)$.

Connections

Links to other subjects: Shift in supply and demand curves (Economics); induced emf and simple harmonic motion (physics).

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AHL content

Recommended teaching hours: 11

The aim of the AHL functions topic is to extend and build upon the aims, concepts and skills from the SL content. It introduces students to useful techniques for finding and using roots of polynomials, graphing and interpreting rational functions, additional ways to classify functions, solving inequations and solving equations involving modulus notation.

HL students may be required to use technology to solve equations where there is no appropriate analytic approach.

AHL 2.12

Content	Guidance, clarification and syllabus links
Polynomial functions, their graphs and equations; zeros, roots and factors. The factor and remainder theorems.	
Sum and product of the roots of polynomial equations.	For the polynomial equation: $\sum_{r=0}^n a_r x^r = 0$, the sum is $\frac{-a_{n-1}}{a_n}$ the product is $\frac{(-1)^n a_0}{a_n}$ Link to: complex roots of quadratic and polynomial equations (AHL 1.14).

Connections

Links to other subjects: Modelling (sciences group subjects)

TOK: Is it an oversimplification to say that some areas of knowledge give us facts whereas other areas of knowledge give us interpretations?

Enrichment: Viète's theorem in full, "The equation that couldn't be solved" quadratic formula reducing a quadratic to a linear, Cardano and Bombelli.

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AHL 2.13

Content	Guidance, clarification and syllabus links
Rational functions of the form $f(x) = \frac{ax + b}{cx^2 + dx + e}$, and $f(x) = \frac{ax^2 + bx + c}{dx + e}$	The reciprocal function is a particular case. Graphs should include all asymptotes (horizontal, vertical and oblique) and any intercepts with axes. Dynamic graphing packages could be used to investigate these functions. Link to: rational functions (SL 2.8).

Connections

International mindedness: Bourbaki group analytical approach versus Mandelbrot visual approach.

TOK: Does studying the graph of a function contain the same level of mathematical rigour as studying the function algebraically? What are the advantages and disadvantages of having different forms and symbolic language in mathematics?

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AHL 2.14

Content	Guidance, clarification and syllabus links
Odd and even functions.	Even: $f(-x) = f(x)$ Odd: $f(-x) = -f(x)$ Includes periodic functions.
Finding the inverse function, $f^{-1}(x)$, including domain restriction.	
Self-inverse functions.	

Connections

International-mindedness: The notation for functions was developed by a number of different mathematicians in the 17th and 18th centuries. How did the notation we use today become internationally accepted?

TOK: If systems of notation and measurement are culturally and historically situated, does this mean mathematics cannot be seen as independent of culture?

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AHL 2.15

Content	Guidance, clarification and syllabus links
Solutions of $g(x) \geq f(x)$, both graphically and analytically.	Graphical or algebraic methods for simple polynomials up to degree 3. Use of technology for these and other functions.

Connections

TOK: Are there differences in terms of value that different cultures ascribe to mathematics, or to the relative value that they ascribe to different areas of knowledge?

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AHL 2.16

Content	Guidance, clarification and syllabus links
The graphs of the functions, $y = f(x) $ and $y = f(x)$, $y = \frac{1}{f(x)}$, $y = f(ax + b)$, $y = [f(x)]^2$.	Dynamic graphing packages could be used to investigate these transformations.
Solution of modulus equations and inequalities.	Example: $ \arccos(x) > 1$

Connections

International-mindedness: The Bourbaki group analytic approach versus Mandelbrot visual approach.

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Topic 3: Geometry and trigonometry

Concepts

Essential understandings:

Geometry and trigonometry allows us to quantify the physical world, enhancing our spatial awareness in two and three dimensions. This topic provides us with the tools for analysis, measurement and transformation of quantities, movements and relationships.

Suggested concepts embedded in this topic:

Generalization, space, relationships, equivalence, representation,

AHL: Quantity, Modelling.

Content-specific conceptual understandings:

- The properties of shapes depend on the dimension they occupy in space.
- Volume and surface area of shapes are determined by formulae, or general mathematical relationships or rules expressed using symbols or variables.
- The relationships between the length of the sides and the size of the angles in a triangle can be used to solve many problems involving position, distance, angles and area.
- Equivalent measurement systems, such as degrees and radians, can be used for angles to facilitate ease of calculation.
- Different representations of the values of trigonometric relationships, such as exact or approximate, may not be equivalent to one another.
- The trigonometric functions of angles may be defined on the unit circle, which can visually and algebraically represent the periodic or symmetric nature of their values.

AHL

- Position and movement can be modelled in three-dimensional space using vectors.
- The relationships between algebraic, geometric and vector methods can help us to solve problems and quantify those positions and movements.

SL content

Recommended teaching hours: 25

The aim of the SL content of the geometry and trigonometry topic is to introduce students to geometry in three dimensions and to non right-angled trigonometry. Students will explore the circular functions and use properties and identities to solve problems in abstract and real-life contexts.

Throughout this topic students should be given the opportunity to use technology such as graphing packages, graphing calculators and dynamic geometry software to develop and apply their knowledge of geometry and trigonometry.

On examination papers, radian measure should be assumed unless otherwise indicated.

Sections SL3.1 to SL3.3 are content common to both Mathematics: analysis and approaches and Mathematics: applications and interpretation.

SL 3.1

Content	Guidance, clarification and syllabus links
The distance between two points in three-dimensional space, and their midpoint.	In SL examinations, only right-angled trigonometry questions will be set in reference to three-dimensional shapes.

Content	Guidance, clarification and syllabus links
Volume and surface area of three-dimensional solids including right-pyramid, right cone, sphere, hemisphere and combinations of these solids. The size of an angle between two intersecting lines or between a line and a plane.	In problems related to these topics, students should be able to identify relevant right-angled triangles in three-dimensional objects and use them to find unknown lengths and angles.

Connections**Other contexts:** Architecture and design.**Links to other subjects:** Design technology; volumes of stars and inverse square law (physics).**TOK:** What is an axiomatic system? Are axioms self evident to everybody?[Download connections template](#)**SL 3.2**

Content	Guidance, clarification and syllabus links
Use of sine, cosine and tangent ratios to find the sides and angles of right-angled triangles.	In all areas of this topic, students should be encouraged to sketch well-labelled diagrams to support their solutions. Link to: inverse functions (SL2.2) when finding angles.
The sine rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$. The cosine rule: $c^2 = a^2 + b^2 - 2ab \cos C$; $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$. Area of a triangle as $\frac{1}{2}ab \sin C$.	This section does not include the ambiguous case of the sine rule.

Connections**Other contexts:** Triangulation, map-making.**Links to other subjects:** Vectors (physics).**International-mindedness:** Diagrams of Pythagoras' theorem occur in early Chinese and Indian manuscripts. The earliest references to trigonometry are in Indian mathematics; the use of triangulation to find the curvature of the Earth in order to settle a dispute between England and France over Newton's gravity.**TOK:** Is it ethical that Pythagoras gave his name to a theorem that may not have been his own creation? What criteria might we use to make such a judgment?[Download connections template](#)**SL 3.3**

Content	Guidance, clarification and syllabus links
Applications of right and non-right angled trigonometry, including Pythagoras's theorem. Angles of elevation and depression.	Contexts may include use of bearings.

Content	Guidance, clarification and syllabus links
Construction of labelled diagrams from written statements.	

Connections

Other contexts: Triangulation, map-making, navigation and radio transmissions. Use of parallax for navigation.

Links to other subjects: Vectors, scalars, forces and dynamics (physics); field studies (sciences group subjects)

Aim 8: Who really invented Pythagoras's theorem?

Aim 9: In how many ways can you prove Pythagoras's theorem?

International-mindedness: The use of triangulation to find the curvature of the Earth in order to settle a dispute between England and France over Newton's gravity.

TOK: If the angles of a triangle can add up to less than 180° , 180° or more than 180° , what does this tell us about the nature of mathematical knowledge?

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SL 3.4

Content	Guidance, clarification and syllabus links
The circle: radian measure of angles; length of an arc; area of a sector.	Radian measure may be expressed as exact multiples of π , or decimals.

Connections

Links to other subjects: Diffraction patterns and circular motion (physics).

International-mindedness: Seki Takakazu calculating π to ten decimal places; Hipparchus, Menelaus and Ptolemy; Why are there 360 degrees in a complete turn? Links to Babylonian mathematics.

TOK: Which is a better measure of angle: radian or degree? What criteria can/do/should mathematicians use to make such decisions?

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SL 3.5

Content	Guidance, clarification and syllabus links
Definition of $\cos\theta$, $\sin\theta$ in terms of the unit circle.	Includes the relationship between angles in different quadrants. $\cos x = \cos(-x)$ Examples: $\tan(3\pi - x) = -\tan x$ $\sin(\pi + x) = -\sin x$
Definition of $\tan\theta$ as $\frac{\sin\theta}{\cos\theta}$.	The equation of a straight line through the origin is $y = x\tan\theta$, where θ is the angle formed between the line and positive x -axis.
Exact values of trigonometric ratios of $0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$ and their multiples.	$\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}, \cos\frac{3\pi}{4} = -\frac{1}{\sqrt{2}}, \tan 210^\circ = \frac{\sqrt{3}}{3}$
Extension of the sine rule to the ambiguous case.	

Connections

International-mindedness: The first work to refer explicitly to the sine as a function of an angle is the Aryabhata of Aryabhata (ca 510).

TOK: Trigonometry was developed by successive civilizations and cultures. To what extent is mathematical knowledge embedded in particular traditions or bound to particular cultures? How have key events in the history of mathematics shaped its current form and methods?

Enrichment: The proof of Pythagoras' theorem in three dimensions.

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SL 3.6

Content	Guidance, clarification and syllabus links
The Pythagorean identity $\cos^2\theta + \sin^2\theta = 1$. Double angle identities for sine and cosine.	Simple geometrical diagrams and dynamic graphing packages may be used to illustrate the double angle identities (and other trigonometric identities).
The relationship between trigonometric ratios.	Examples: Given $\sin\theta$, find possible values of $\tan\theta$, (without finding θ). Given $\cos x = \frac{3}{4}$ and x is acute, find $\sin 2x$, (without finding x).

Connections

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SL 3.7

Content	Guidance, clarification and syllabus links
The circular functions $\sin x$, $\cos x$, and $\tan x$; amplitude, their periodic nature, and their graphs Composite functions of the form $f(x) = a\sin(b(x+c)) + d$.	Trigonometric functions may have domains given in degrees or radians. Examples: $f(x) = \tan(x - \frac{\pi}{4})$, $f(x) = 2\cos(3(x-4)) + 1$.
Transformations.	Example: $y = \sin x$ used to obtain $y = 3\sin 2x$ by a stretch of scale factor 3 in the y direction and a stretch of scale factor $\frac{1}{2}$ in the x direction. Link to: transformations of graphs (SL2.11).
Real-life contexts.	Examples: height of tide, motion of a Ferris wheel. Students should be aware that not all regression technology produces trigonometric functions in the form $f(x) = a\sin(b(x+c)) + d$.

Connections

Links to other subjects: Simple harmonic motion (physics).

TOK: Music can be expressed using mathematics. What does this tell us about the relationship between music and mathematics?

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SL 3.8

Content	Guidance, clarification and syllabus links
Solving trigonometric equations in a finite interval, both graphically and analytically.	$2\sin x = 1, \quad 0 \leq x \leq 2\pi$ Examples: $2\sin 2x = 3\cos x, \quad 0^\circ \leq x \leq 180^\circ$ $2\tan(3(x-4)) = 1, \quad -\pi \leq x \leq 3\pi$
Equations leading to quadratic equations in $\sin x$, $\cos x$ or $\tan x$.	Examples: $2\sin^2 x + 5\cos x + 1 = 0$ for $0 \leq x \leq 4\pi$, $2\sin x = \cos 2x, \quad -\pi \leq x \leq \pi$ Not required: The general solution of trigonometric equations.

Connections
[Download connections template](#)
AHL content

Recommended teaching hours: 26

The aim of the AHL content in the geometry and trigonometry topic is to extend and build upon the aims, concepts and skills from the SL content. It further explores the circular functions, introduces some important trigonometric identities, and introduces vectors in two and three dimensions. This will facilitate problem-solving involving points, lines and planes.

On examination papers radian measure should be assumed unless otherwise indicated.

AHL 3.9

Content	Guidance, clarification and syllabus links
Definition of the reciprocal trigonometric ratios $\sec \theta$, $\operatorname{cosec} \theta$ and $\cot \theta$. Pythagorean identities: $1 + \tan^2 \theta = \sec^2 \theta$ $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$ The inverse functions $f(x) = \arcsin x$, $f(x) = \arccos x$, $f(x) = \arctan x$; their domains and ranges; their graphs.	

Connections

International-mindedness: The origin of degrees in the mathematics of Mesopotamia and why we use minutes and seconds for time; the origin of the word sine.

TOK: What is the relationship between concepts and facts? To what extent do the concepts that we use shape the conclusions that we reach?

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AHL 3.10

Content	Guidance, clarification and syllabus links
Compound angle identities. Double angle identity for tan.	Derivation of double angle identities from compound angle identities. Link to: De Moivre's theorem (AHL1.14).

Connections

Other contexts: Triangulation used by GPSs (global positioning systems); concepts in electrical engineering including generation of sinusoidal voltage.

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AHL 3.11

Content	Guidance, clarification and syllabus links
Relationships between trigonometric functions and the symmetry properties of their graphs.	$\sin(\pi - \theta) = \sin \theta$ $\cos(\pi - \theta) = -\cos \theta$ $\tan(\pi - \theta) = -\tan \theta$ Link to: the unit circle (SL3.5), odd and even functions (AHL2.14), compound angles (AHL3.10).

Connections

Links to other subjects: Simple harmonic motion graphs (physics)

TOK: Mathematics and knowledge claims: how can there be an infinite number of discrete solutions to an equation?

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AHL 3.12

Content	Guidance, clarification and syllabus links
Concept of a vector; position vectors; displacement vectors. Representation of vectors using directed line segments. Base vectors i , j , k . Components of a vector: $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}.$	
Algebraic and geometric approaches to the following: - the sum and difference of two vectors - the zero vector $\mathbf{0}$, the vector $-\mathbf{v}$ - multiplication by a scalar, $k\mathbf{v}$, parallel vectors	Distance between points A and B is the magnitude of $\vec{AB}$

Content	Guidance, clarification and syllabus links
• magnitude of a vector, v; unit vectors, $\frac{v}{ v }$ • position vectors $\vec{OA} = a$, $\vec{OB} = b$ • displacement vector $\vec{AB} = b - a$ Proofs of geometrical properties using vectors.	

Connections

Links to other subjects: Vectors, scalars, forces and dynamics (physics).

Aim 8: Vectors are used to solve many problems in position location. This can be used to save a lost sailor or destroy a building with a laser-guided bomb.

TOK: Vectors are used to solve many problems in position location. This can be used to save a lost sailor or destroy a building with a laser-guided bomb. To what extent does possession of knowledge carry with it an ethical obligation?

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AHL 3.13

Content	Guidance, clarification and syllabus links
The definition of the scalar product of two vectors. The angle between two vectors. Perpendicular vectors; parallel vectors.	Applications of the properties of the scalar product $v \cdot w = w \cdot v$; $u \cdot (v + w) = u \cdot v + u \cdot w$; $(kv) \cdot w = k(v \cdot w)$; $v \cdot v = v ^2$. $v \cdot w = v w \cos\theta$, where θ is the angle between v and w . For non-zero vectors $v \cdot w = 0$ is equivalent to the vectors being perpendicular; for parallel vectors $ v \cdot w = v w $.

Connections

Links to other subjects: Forces and dynamics (physics).

TOK: The nature of mathematics: why this definition of scalar product?

Enrichment: Proof of the cosine rule using the dot product.

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AHL 3.14

Content	Guidance, clarification and syllabus links
Vector equation of a line in two and three dimensions: $r = a + \lambda b$.	Relevance of a (position) and b (direction). Knowledge of the following forms for equations of lines: Parametric form: $x = x_0 + \lambda l$, $y = y_0 + \lambda m$, $z = z_0 + \lambda n$.

Content	Guidance, clarification and syllabus links
	Cartesian form: $\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}.$
The angle between two lines.	Using the scalar product of the two direction vectors.
Simple applications to kinematics.	Interpretation of λ as time and $\mathbf{b}$ as velocity, with $ \mathbf{b} $ representing speed.

Connections

Other contexts: Modelling linear motion in three dimensions; navigational devices, for example GPS.

TOK: Why might it be argued that one form of representation is superior to another? What criteria might a mathematician use in making such an argument?

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AHL 3.15

Content	Guidance, clarification and syllabus links
Coincident, parallel, intersecting and skew lines, distinguishing between these cases. Points of intersection.	Skew lines are non-parallel lines that do not intersect in three-dimensional space.

Connections

TOK: How can there be an infinite number of discrete solutions to an equation? What does this suggest about the nature of mathematical knowledge and how it compares to knowledge in other disciplines?

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AHL 3.16

Content	Guidance, clarification and syllabus links
The definition of the vector product of two vectors.	The vector product is also known as the “cross product”. $\mathbf{v} \times \mathbf{w} = \mathbf{v} \mathbf{w} \sin\theta\mathbf{n}$, where θ is the angle between $\mathbf{v}$ and $\mathbf{w}$, and $\mathbf{n}$ is the unit normal vector whose direction is given by the right-hand screw rule.
Properties of the vector product.	$\mathbf{v} \times \mathbf{w} = -\mathbf{w} \times \mathbf{v}$; $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$; $(k\mathbf{v}) \times \mathbf{w} = k(\mathbf{v} \times \mathbf{w})$; $\mathbf{v} \times \mathbf{v} = \mathbf{0}$. For non-zero vectors $\mathbf{v} \times \mathbf{w} = \mathbf{0}$ is equivalent to the vectors being parallel.
Geometric interpretation of $ \mathbf{v} \times \mathbf{w} $	Use of $ \mathbf{v} \times \mathbf{w} $ to find the area of a parallelogram (and hence a triangle).

Connections

Links to other subjects: Magnetic forces and fields (physics).

TOK: To what extent is certainty attainable in mathematics? Is certainty attainable, or desirable, in other areas of knowledge?

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AHL 3.17

Content	Guidance, clarification and syllabus links
Vector equations of a plane: $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b} + \mu \mathbf{c}$, where $\mathbf{b}$ and $\mathbf{c}$ are non-parallel vectors within the plane. $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$, where $\mathbf{n}$ is a normal to the plane and $\mathbf{a}$ is the position vector of a point on the plane. Cartesian equation of a plane $ax + by + cz = d$.	

Connections

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AHL 3.18

Content	Guidance, clarification and syllabus links
Intersections of: a line with a plane; two planes; three planes. Angle between: a line and a plane; two planes.	Finding intersections by solving equations; geometrical interpretation of solutions. Link to: solutions of systems of linear equations (AHL 1.16).

Connections

TOK: Mathematics and the knower: are symbolic representations of three-dimensional objects easier to deal with than visual representations? What does this tell us about our knowledge of mathematics in other dimensions?

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Topic 4: Statistics and probability

Concepts

Essential understandings:

Statistics is concerned with the collection, analysis and interpretation of data and the theory of probability can be used to estimate parameters, discover empirical laws, test hypotheses and predict the occurrence of events. Statistical representations and measures allow us to represent data in many different forms to aid interpretation.

Probability enables us to quantify the likelihood of events occurring and so evaluate risk. Both statistics and probability provide important representations which enable us to make predictions, valid comparisons and informed decisions. These fields have power and limitations and should be applied with care and critically

questioned to differentiate between the theoretical and the empirical/observed. Probability theory allows us to make informed choices, to evaluate risk, and to make predictions about seemingly random events.

Suggested concepts embedded in this topic:

Quantity, validity, approximation, generalization.

AHL: Change, systems.

Content-specific conceptual understandings:

- Organizing, representing, analysing and interpreting data and utilizing different statistical tools facilitates prediction and drawing of conclusions.
- Different statistical techniques require justification and the identification of their limitations and validity.
- Approximation in data can approach the truth but may not always achieve it.
- Some techniques of statistical analysis, such as regression, standardization or formulae, can be applied in a practical context to apply to general cases.
- Modelling through statistics can be reliable, but may have limitations.

AHL

- Properties of probability density functions can be used to identify measure of central tendency such as mean, mode and median.
- Probability methods such as Bayes theorem can be applied to real-world systems, such as medical studies or economics, to inform decisions and to better understand outcomes.

SL content

Recommended teaching hours: 27

The aim of the SL content in the statistics and probability topic is to introduce students to the important concepts, techniques and representations used in statistics and probability. Students should be given the opportunity to approach this topic in a practical way, to understand why certain techniques are used and to interpret the results. The use of technology such as simulations, spreadsheets, statistics software and statistics apps can greatly enhance this topic.

It is expected that most of the calculations required will be carried out using technology, but explanations of calculations by hand may enhance understanding. The emphasis is on understanding and interpreting the results obtained, in context.

In examinations students should be familiar with how to use the statistics functionality of allowed technology.

At SL the data set will be considered to be the population unless otherwise stated.

Sections SL4.1 to SL4.9 are content common to both Mathematics: analysis and approaches and Mathematics: applications and interpretation.

SL 4.1

Content	Guidance, clarification and syllabus links
Concepts of population, sample, random sample, discrete and continuous data.	This is designed to cover the key questions that students should ask when they see a data set/ analysis.
Reliability of data sources and bias in sampling.	Dealing with missing data, errors in the recording of data.
Interpretation of outliers.	Outlier is defined as a data item which is more than $1.5 \times$ interquartile range (IQR) from the nearest quartile.

Content	Guidance, clarification and syllabus links
	Awareness that, in context, some outliers are a valid part of the sample but some outlying data items may be an error in the sample. Link to: box and whisker diagrams (SL4.2) and measures of dispersion (SL4.3).
Sampling techniques and their effectiveness.	Simple random, convenience, systematic, quota and stratified sampling methods.

Connections

Links to other subjects: Descriptive statistics and random samples (biology, psychology, sports exercise and health science, environmental systems and societies, geography, economics; business management); research methodologies (psychology).

Aim 8: Misleading statistics; examples of problems caused by absence of representative samples, for example Google flu predictor, US presidential elections in 1936, Literary Digest v George Gallup, Boston “pot-hole” app.

International-mindedness: The Kinsey report—famous sampling techniques.

TOK: Why have mathematics and statistics sometimes been treated as separate subjects? How easy is it to be misled by statistics? Is it ever justifiable to purposely use statistics to mislead others?

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SL 4.2

Content	Guidance, clarification and syllabus links
Presentation of data (discrete and continuous): frequency distributions (tables).	Class intervals will be given as inequalities, without gaps.
Histograms. Cumulative frequency; cumulative frequency graphs; use to find median, quartiles, percentiles, range and interquartile range (IQR).	Frequency histograms with equal class intervals. Not required: Frequency density histograms.
Production and understanding of box and whisker diagrams.	Use of box and whisker diagrams to compare two distributions, using symmetry, median, interquartile range or range. Outliers should be indicated with a cross. Determining whether the data may be normally distributed by consideration of the symmetry of the box and whiskers.

Connections

Links to other subjects: Presentation of data (sciences, individuals and societies).

International-mindedness: Discussion of the different formulae for the same statistical measure (for example, variance).

TOK: What is the difference between information and data? Does “data” mean the same thing in different areas of knowledge?

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SL 4.3

Content	Guidance, clarification and syllabus links
Measures of central tendency (mean, median and mode). Estimation of mean from grouped data.	Calculation of mean using formula and technology. Students should use mid-interval values to estimate the mean of grouped data.
Modal class.	For equal class intervals only.
Measures of dispersion (interquartile range, standard deviation and variance).	Calculation of standard deviation and variance of the sample using only technology, however hand calculations may enhance understanding. Variance is the square of the standard deviation.
Effect of constant changes on the original data.	Examples: If three is subtracted from the data items, then the mean is decreased by three, but the standard deviation is unchanged. If all the data items are doubled, the mean is doubled and the standard deviation is also doubled.
Quartiles of discrete data.	Using technology. Awareness that different methods for finding quartiles exist and therefore the values obtained using technology and by hand may differ.

Connections

Other contexts: Comparing variation and spread in populations, human or natural, for example agricultural crop data, social indicators, reliability and maintenance.

Links to other subjects: Descriptive statistics (sciences and individuals and societies); consumer price index (economics).

International-mindedness: The benefits of sharing and analysing data from different countries; discussion of the different formulae for variance.

TOK: Could mathematics make alternative, equally true, formulae? What does this tell us about mathematical truths? Does the use of statistics lead to an over-emphasis on attributes that can be easily measured over those that cannot?

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SL 4.4

Content	Guidance, clarification and syllabus links
Linear correlation of bivariate data. Pearson's product-moment correlation coefficient, r .	Technology should be used to calculate r . However, hand calculations of r may enhance understanding. Critical values of r will be given where appropriate. Students should be aware that Pearson's product moment correlation coefficient (r) is only meaningful for linear relationships.
Scatter diagrams; lines of best fit, by eye, passing through the mean point.	Positive, zero, negative; strong, weak, no correlation. Students should be able to make the distinction between correlation and causation and know that correlation does not imply causation.
Equation of the regression line of y on x .	Technology should be used to find the equation.

Content	Guidance, clarification and syllabus links
Use of the equation of the regression line for prediction purposes. Interpret the meaning of the parameters, a and b , in a linear regression $y = ax + b$.	Students should be aware: • of the dangers of extrapolation • that they cannot always reliably make a prediction of x from a value of y, when using a y on x line.

Connections

Other contexts: Linear regressions where correlation exists between two variables. Exploring cause and dependence for categorical variables, for example, on what factors might political persuasion depend?

Links to other subjects: Curves of best fit, correlation and causation (sciences group subjects); scatter graphs (geography).

Aim 8: The correlation between smoking and lung cancer was “discovered” using mathematics. Science had to justify the cause.

TOK: Correlation and causation—can we have knowledge of cause and effect relationships given that we can only observe correlation? What factors affect the reliability and validity of mathematical models in describing real-life phenomena?

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SL 4.5

Content	Guidance, clarification and syllabus links
Concepts of trial, outcome, equally likely outcomes, relative frequency, sample space (U) and event. The probability of an event A is $P(A) = \frac{n(A)}{n(U)}$. The complementary events A and A' (not A).	Sample spaces can be represented in many ways, for example as a table or a list. Experiments using coins, dice, cards and so on, can enhance understanding of the distinction between experimental (relative frequency) and theoretical probability. Simulations may be used to enhance this topic.
Expected number of occurrences.	Example: If there are 128 students in a class and the probability of being absent is 0.1, the expected number of absent students is 12.8.

Connections

Other contexts: Actuarial studies and the link between probability of life spans and insurance premiums, government planning based on likely projected figures, Monte Carlo methods.

Links to other subjects: Theoretical genetics and Punnett squares (biology); the position of a particle (physics).

Aim 8: The ethics of gambling.

International-mindedness: The St Petersburg paradox; Chebyshev and Pavlovsky (Russian).

TOK: To what extent are theoretical and experimental probabilities linked? What is the role of emotion in our perception of risk, for example in business, medicine and travel safety?

Use of technology: Computer simulations may be useful to enhance this topic.

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SL 4.6

Content	Guidance, clarification and syllabus links
Use of Venn diagrams, tree diagrams, sample space diagrams and tables of outcomes to calculate probabilities.	
Combined events: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Mutually exclusive events: $P(A \cap B) = 0$.	The non-exclusivity of “or”.
Conditional probability: $P(A B) = \frac{P(A \cap B)}{P(B)}$.	An alternate form of this is: $P(A \cap B) = P(B)P(A B)$. Problems can be solved with the aid of a Venn diagram, tree diagram, sample space diagram or table of outcomes without explicit use of formulae. Probabilities with and without replacement.
Independent events: $P(A \cap B) = P(A)P(B)$.	

Connections

Aim 8: The gambling issue: use of probability in casinos. Could or should mathematics help increase incomes in gambling?

TOK: Can calculation of gambling probabilities be considered an ethical application of mathematics? Should mathematicians be held responsible for unethical applications of their work?

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SL 4.7

Content	Guidance, clarification and syllabus links												
Concept of discrete random variables and their probability distributions. Expected value (mean), for discrete data. Applications.	Probability distributions will be given in the following ways: <table><tr><td>X</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$P(X = x)$</td><td>0.1</td><td>0.2</td><td>0.15</td><td>0.05</td><td>0.5</td></tr></table> $P(X = x) = \frac{1}{18}(4 + x) \text{ for } x \in \{1, 2, 3\}$ $E(X) = 0$ indicates a fair game where X represents the gain of a player.	X	1	2	3	4	5	$P(X = x)$	0.1	0.2	0.15	0.05	0.5
X	1	2	3	4	5								
$P(X = x)$	0.1	0.2	0.15	0.05	0.5								

Connections

Other contexts: Games of chance.

Aim 8: Why has it been argued that theories based on the calculable probabilities found in casinos are pernicious when applied to everyday life (for example, economics)?

TOK: What do we mean by a “fair” game? Is it fair that casinos should make a profit?

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SL 4.8

Content	Guidance, clarification and syllabus links
Binomial distribution. Mean and variance of the binomial distribution.	Situations where the binomial distribution is an appropriate model. In examinations, binomial probabilities should be found using available technology. Not required: Formal proof of mean and variance. Link to: expected number of occurrences (SL4.5).

Connections

Aim 8: Pascal's triangle, attributing the origin of a mathematical discovery to the wrong mathematician.

International-mindedness: The so-called "Pascal's triangle" was known to the Chinese mathematician Yang Hui much earlier than Pascal.

TOK: What criteria can we use to decide between different models?

Enrichment: Hypothesis testing using the binomial distribution.

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SL 4.9

Content	Guidance, clarification and syllabus links
The normal distribution and curve. Properties of the normal distribution. Diagrammatic representation.	Awareness of the natural occurrence of the normal distribution. Students should be aware that approximately 68% of the data lies between $\mu \pm \sigma$, 95% lies between $\mu \pm 2\sigma$ and 99.7% of the data lies between $\mu \pm 3\sigma$.
Normal probability calculations.	Probabilities and values of the variable must be found using technology.
Inverse normal calculations	For inverse normal calculations mean and standard deviation will be given. This does not involve transformation to the standardized normal variable z .

Connections

Links to other subjects: Normally distributed real-life measurements and descriptive statistics (sciences group subjects, psychology, environmental systems and societies)

Aim 8: Why might the misuse of the normal distribution lead to dangerous inferences and conclusions?

International-mindedness: De Moivre's derivation of the normal distribution and Quetelet's use of it to describe *l'homme moyen*.

TOK: To what extent can we trust mathematical models such as the normal distribution? How can we know what to include, and what to exclude, in a model?

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SL 4.10

Content	Guidance, clarification and syllabus links
Equation of the regression line of x on y .	
Use of the equation for prediction purposes.	Students should be aware that they cannot always reliably make a prediction of y from a value of x , when using an x on y line.

Connections

TOK: Is it possible to have knowledge of the future?

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SL 4.11

Content	Guidance, clarification and syllabus links
Formal definition and use of the formulae: $P(A B) = \frac{P(A \cap B)}{P(B)}$ for conditional probabilities, and $P(A B) = P(A) = P(A B')$ for independent events.	An alternate form of this is: $P(A \cap B) = P(B)P(A B)$. Testing for independence.

Connections

Other contexts: Use of probability methods in medical studies to assess risk factors for certain diseases.

TOK: Given the interdisciplinary nature of many real-world applications of probability, is the division of knowledge into discrete disciplines or areas of knowledge artificial and/or useful?

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SL 4.12

Content	Guidance, clarification and syllabus links
Standardization of normal variables (z -values).	Probabilities and values of the variable must be found using technology. The standardized value (z) gives the number of standard deviations from the mean.
Inverse normal calculations where mean and standard deviation are unknown.	Use of z -values to calculate unknown means and standard deviations.

Connections

Links to other subjects: The normal distribution (biology); descriptive statistics (psychology).

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AHL content

Recommended teaching hours: 6

The aim of the AHL content in the statistics and probability topic is to extend and build upon the aims, concepts and skills from the SL content. Students are introduced to further conditional probability theory in

the form of Bayes Theorem and properties of discrete and continuous random variables are further explored.

AHL 4.13

Content	Guidance, clarification and syllabus links
Use of Bayes' theorem for a maximum of three events.	Link to: independent events (SL4.6).

Connections

Other contexts: Use of probability methods in medical studies to assess risk factors for certain diseases.

TOK: Does the applicability of knowledge vary across the different areas of knowledge? What would the implications be if the value of all knowledge was measured solely in terms of its applicability?

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AHL 4.14

Content	Guidance, clarification and syllabus links
Variance of a discrete random variable.	Link to: discrete random variables (SL 4.7)
Continuous random variables and their probability density functions.	$0 \leq f(x) \leq 1$, $\int_{-\infty}^{\infty} f(x)dx = 1$ including piecewise functions.
Mode and median of continuous random variables.	For a continuous random variable, a value at which the probability density function has a maximum value is called a mode and for the median: $\int_{-\infty}^m f(x)dx = \frac{1}{2}$.
Mean, variance and standard deviation of both discrete and continuous random variables.	Use of the notation $E(X)$, $E(X^2)$, $\text{Var}(X)$, where $\text{Var}(X) = E(X^2) - [E(X)]^2$ and related formulae. Use of $E(X)$ for "fair" games.
The effect of linear transformations of X .	$E(aX + b) = aE(X) + b$, $\text{Var}(aX + b) = a^2\text{Var}(X)$

Connections

Other contexts: Other discrete distributions, for example Poisson, may be appropriate for IA/toolkit and further investigation; expected value used in decision making in business, economics and life in general; expected gain to insurance companies.

TOK: Is mathematics more or less useful than other areas of knowledge for solving problems?

Enrichment: Is there a relationship between the interquartile range and the standard deviation for a normally distributed data set?

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Topic 5: Calculus

Concepts

Essential understandings:

Calculus describes rates of change between two variables and the accumulation of limiting areas. Understanding these rates of change and accumulations allow us to model, interpret and analyze real-world problems and situations. Calculus helps us to understand the behaviour of functions and allows us to interpret the features of their graphs.

Suggested concepts embedded in this topic:

Change, patterns, relationships, approximation, generalization, space, modelling.

AHL: Systems, quantity.

Content-specific conceptual understandings:

- The derivative may be represented physically as a rate of change and geometrically as the gradient or slope function.
- Areas under curves can be approximated by the sum of the areas of rectangles which may be calculated even more accurately using integration.
- Examining rates of change close to turning points helps to identify intervals where the function increases/decreases, and identify the concavity of the function.
- Numerical integration can be used to approximate areas in the physical world.
- Mathematical modelling can provide effective solutions to real-life problems in optimization by maximizing or minimizing a quantity, such as cost or profit.
- Derivatives and integrals describe real-world kinematics problems in two and three-dimensional space by examining displacement, velocity and acceleration.

AHL

- Some functions may be continuous everywhere but not differentiable everywhere.
- A finite number of terms of an infinite series can be a general approximation of a function over a limited domain.
- Limits describe the output of a function as the input approaches a certain value and can represent convergence and divergence.
- Examining limits of functions at a point can help determine continuity and differentiability at a point.

SL content

Recommended teaching hours: 28

The aim of the SL content in the calculus topic is to introduce students to the concepts and techniques of differential and integral calculus and their applications.

Throughout this topic students should be given the opportunity to use technology such as graphing packages and graphing calculators to develop and apply their knowledge of calculus.

Sections SL5.1 to SL5.5 are content common to both Mathematics: analysis and approaches and Mathematics: applications and interpretation.

SL 5.1

Content	Guidance, clarification and syllabus links
Introduction to the concept of a limit.	Estimation of the value of a limit from a table or graph.

Content	Guidance, clarification and syllabus links
	Not required: Formal analytic methods of calculating limits.
Derivative interpreted as gradient function and as rate of change.	Forms of notation: $\frac{dy}{dx}$, $f'(x)$, $\frac{dV}{dr}$ or $\frac{ds}{dt}$ for the first derivative. Informal understanding of the gradient of a curve as a limit.

Connections

Links to other subjects: Marginal cost, marginal revenue, marginal profit, market structures (economics); kinematics, induced emf and simple harmonic motion (physics); interpreting the gradient of a curve (chemistry)

Aim 8: The debate over whether Newton or Leibnitz discovered certain calculus concepts; how the Greeks' distrust of zero meant that Archimedes' work did not lead to calculus.

International-mindedness: Attempts by Indian mathematicians (500-1000 CE) to explain division by zero.

TOK: What value does the knowledge of limits have? Is infinitesimal behaviour applicable to real life? Is intuition a valid way of knowing in mathematics?

Use of technology: Spreadsheets, dynamic graphing software and GDC should be used to explore ideas of limits, numerically and graphically. Hypotheses can be formed and then tested using technology.

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SL 5.2

Content	Guidance, clarification and syllabus links
Increasing and decreasing functions. Graphical interpretation of $f'(x) > 0$, $f'(x) = 0$, $f'(x) < 0$.	Identifying intervals on which functions are increasing ($f'(x) > 0$) or decreasing ($f'(x) < 0$).

Connections

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SL 5.3

Content	Guidance, clarification and syllabus links
Derivative of $f(x) = ax^n$ is $f'(x) = anx^{n-1}$, $n \in \mathbb{Z}$ The derivative of functions of the form $f(x) = ax^n + bx^{n-1} \dots$ where all exponents are integers.	

Connections

TOK: The seemingly abstract concept of calculus allows us to create mathematical models that permit human feats such as getting a man on the Moon. What does this tell us about the links between mathematical models and reality?

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SL 5.4

Content	Guidance, clarification and syllabus links
Tangents and normals at a given point, and their equations.	Use of both analytic approaches and technology.

Connections

Links to other subjects: Instantaneous velocity and optics, equipotential surfaces (physics); price elasticity (economics).

TOK: In what ways has technology impacted how knowledge is produced and shared in mathematics? Does technology simply allow us to arrange existing knowledge in new and different ways, or should this arrangement itself be considered knowledge?

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SL 5.5

Content	Guidance, clarification and syllabus links
Introduction to integration as anti-differentiation of functions of the form $f(x) = ax^n + bx^{n-1} + \dots$, where $n \in \mathbb{Z}$, $n \neq -1$	Students should be aware of the link between anti-derivatives, definite integrals and area.
Anti-differentiation with a boundary condition to determine the constant term.	Example: If $\frac{dy}{dx} = 3x^2 + x$ and $y = 10$ when $x = 1$, then $y = x^3 + \frac{1}{2}x^2 + 8.5$.
Definite integrals using technology. Area of a region enclosed by a curve $y = f(x)$ and the x -axis, where $f(x) > 0$.	Students are expected to first write a correct expression before calculating the area, for example $\int_2^6 (3x^2 + 4)dx$. The use of dynamic geometry or graphing software is encouraged in the development of this concept.

Connections

Other contexts: Velocity-time graphs

Links to other subjects: Velocity-time and acceleration-time graphs (physics and sports exercise and health science)

TOK: Is it possible for an area of knowledge to describe the world without transforming it?

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SL 5.6

Content	Guidance, clarification and syllabus links
Derivative of x^n ($n \in \mathbb{Q}$), $\sin x$, $\cos x$, e^x and $\ln x$. Differentiation of a sum and a multiple of these functions.	
The chain rule for composite functions.	Example: $f(x) = e^{(x^2 + 2)}$, $f(x) = \sin(3x - 1)$

Content	Guidance, clarification and syllabus links
The product and quotient rules.	Link to: composite functions (SL2.5).

Connections

Links to other subjects: Uniform circular motion and induced emf (physics).

TOK: What is the role of convention in mathematics? Is this similar or different to the role of convention in other areas of knowledge?

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SL 5.7

Content	Guidance, clarification and syllabus links
The second derivative. Graphical behaviour of functions, including the relationship between the graphs of f , f' and f'' .	Use of both forms of notation, $\frac{d^2y}{dx^2}$ and $f''(x)$. Technology can be used to explore graphs and calculate the derivatives of functions. Link to: function graphing skills (SL2.3).

Connections

Links to other subjects: Simple harmonic motion (physics).

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SL 5.8

Content	Guidance, clarification and syllabus links
Local maximum and minimum points. Testing for maximum and minimum.	Using change of sign of the first derivative or using sign of the second derivative where $f''(x) > 0$ implies a minimum and $f''(x) < 0$ implies a maximum.
Optimization.	Examples of optimization may include profit, area and volume.
Points of inflexion with zero and non-zero gradients.	At a point of inflexion, $f''(x) = 0$ and changes sign (concavity change), for example $f''(x) = 0$ is not a sufficient condition for a point of inflexion for $y = x^4$ at $(0, 0)$. Use of the terms "concave-up" for $f''(x) > 0$, and "concave-down" for $f''(x) < 0$.

Connections

Other contexts: Profit, area, volume.

Links to other subjects: Velocity-time graphs, simple harmonic motion graphs and kinematics (physics); allocative efficiency (economics).

TOK: When mathematicians and historians say that they have explained something, are they using the word "explain" in the same way?

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SL 5.9

Content	Guidance, clarification and syllabus links
Kinematic problems involving displacement s , velocity v , acceleration a and total distance travelled.	$v = \frac{ds}{dt}; \quad a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$ Displacement from t_1 to t_2 is given by $\int_{t_1}^{t_2} v(t)dt$. Distance between t_1 to t_2 is given by $\int_{t_1}^{t_2} v(t) dt$. Speed is the magnitude of velocity.

Connections

Links to other subjects: Kinematics (physics).

International-mindedness: Does the inclusion of kinematics as core mathematics reflect a particular cultural heritage? Who decides what is mathematics?

TOK: Is mathematics independent of culture? To what extent are we people aware of the impact of culture on what we they believe or know?

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SL 5.10

Content	Guidance, clarification and syllabus links
Indefinite integral of x^n ($n \in \mathbb{Q}$), $\sin x$, $\cos x$, $\frac{1}{x}$ and e^x .	$\int \frac{1}{x} dx = \ln x + C$
The composites of any of these with the linear function $ax + b$.	Example: $f'(x) = \cos(2x + 3) \Rightarrow f(x) = \frac{1}{2}\sin(2x + 3) + C$
Integration by inspection (reverse chain rule) or by substitution for expressions of the form: $\int kg'(x)f(g(x))dx$.	Examples: $\int 2x(x^2 + 1)^4 dx, \int 4x \sin x^2 dx, \int \frac{\sin x}{\cos x} dx.$

Connections

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SL 5.11

Content	Guidance, clarification and syllabus links
Definite integrals, including analytical approach.	$\int_a^b g'(x)dx = g(b) - g(a).$ The value of some definite integrals can only be found using technology. Link to: definite integrals using technology (SL5.5).

Content	Guidance, clarification and syllabus links
Areas of a region enclosed by a curve $y = f(x)$ and the x -axis, where $f(x)$ can be positive or negative, without the use of technology. Areas between curves.	Students are expected to first write a correct expression before calculating the area. Technology may be used to enhance understanding of the relationship between integrals and areas.

Connections

International-mindedness: Accurate calculation of the volume of a cylinder by Chinese mathematician Liu Hui; Ibn Al Haytham: first mathematician to calculate the integral of a function, in order to find the volume of a paraboloid.

TOK: Consider $f(x) = \frac{1}{x}$, $1 \leq x \leq \infty$. An infinite area sweeps out a finite volume. Can this be reconciled with our intuition? Do emotion and intuition have a role in mathematics?

Enrichment: Exploring numerical integration techniques such as Simpson's rule or the trapezoidal rule.

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AHL content

Recommended teaching hours: 27

The aim of the AHL content in the calculus topic is to extend and build upon the aims, concepts and skills from the SL content. Further powerful techniques and useful applications of differential and integral calculus are introduced.

AHL 5.12

Content	Guidance, clarification and syllabus links
Informal understanding of continuity and differentiability of a function at a point.	In examinations, students will not be asked to test for continuity and differentiability.
Understanding of limits (convergence and divergence). Definition of derivative from first principles $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$	Link to: infinite geometric sequences (SL 1.8). Use of this definition for polynomials only.
Higher derivatives.	Familiarity with the notations $\frac{d^ny}{dx^n}$, $f^{(n)}(x)$. Link to: proof by mathematical induction (AHL 1.15).

Connections

Links to other subjects: Theory of the firm (economics).

International-mindedness: How the Greeks' distrust of zero meant that Archimedes' work did not lead to the Calculus; investigate attempts by Indian mathematicians (500-1000AD) to explain division by zero.

TOK: Does the fact that Leibniz and Newton came across the Calculus at similar times support the argument of Platonists over Constructivists?

Enrichment: Fundamental theorem of calculus.

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AHL 5.13

Content	Guidance, clarification and syllabus links
The evaluation of limits of the form $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ and $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$ using l'Hôpital's rule or the Maclaurin series.	The indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$. For example: $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$. Link to: horizontal asymptotes (SL2.8) .
Repeated use of l'Hôpital's rule.	

Connections

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AHL 5.14

Content	Guidance, clarification and syllabus links
Implicit differentiation. Related rates of change. Optimisation problems.	Appropriate use of the chain rule or implicit differentiation, including cases where the optimum solution is at the end point.

Connections

Other contexts: Links between mathematical and physical models.

TOK: Euler was able to make important advances in mathematical analysis before calculus had been put on a solid theoretical foundation by Cauchy and others. However, some work was not possible until after Cauchy's work. What does this suggest about the nature of progress and development in mathematics? How might this be similar/different to the nature of progress and development in other areas of knowledge?

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AHL 5.15

Content	Guidance, clarification and syllabus links
Derivatives of $\tan x$, $\sec x$, $\operatorname{cosec} x$, $\cot x$, a^x , $\log_a x$, $\arcsin x$, $\arccos x$, $\arctan x$.	
Indefinite integrals of the derivatives of any of the above functions. The composites of any of these with a linear function.	Indefinite integral interpreted as a family of curves. Examples: $\int \frac{1}{x^2 + 2x + 5} dx = \frac{1}{2} \arctan \frac{(x+1)}{2} + C$ $\int \sec^2(2x + 5) dx = \frac{1}{2} \tan(2x + 5) + C$
Use of partial fractions to rearrange the integrand.	$\int \frac{1}{x^2 + 3x + 2} dx = \ln \left \frac{x+1}{x+2} \right + C$ Link to: partial fractions (AHL1.11)

Connections

TOK: Can a mathematical statement be true before it has been proven?

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AHL 5.16

Content	Guidance, clarification and syllabus links
Integration by substitution.	On examination papers, substitutions will be provided if the integral is not of the form $\int kg'(x)f(g(x))dx$. Link to: integration by substitution (SL5.10).
Integration by parts.	Examples: $\int x \sin x dx$, $\int \ln x dx$, $\int \arcsin x dx$
Repeated integration by parts.	Examples: $\int x^2 e^x dx$ and $\int e^x \sin x dx$.

Connections

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AHL 5.17

Content	Guidance, clarification and syllabus links
Area of the region enclosed by a curve and the y -axis in a given interval. Volumes of revolution about the x -axis or y -axis.	

Connections

Other contexts: Industrial design.

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AHL 5.18

Content	Guidance, clarification and syllabus links
First order differential equations. Numerical solution of $\frac{dy}{dx} = f(x, y)$ using Euler's method.	$x_{n+1} = x_n + h$, where h is a constant.
Variables separable.	<i>Example:</i> the logistic equation $\frac{dn}{dt} = kn(a - n)$, $a, k \in \mathbb{R}$ Link to: partial fractions (AHL1.11) and use of partial fractions to rearrange the integrand (AHL5.15).
Homogeneous differential equation $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ using the substitution $y = vx$.	

Content	Guidance, clarification and syllabus links
Solution of $y' + P(x)y = Q(x)$, using the integrating factor.	

Connections

Other contexts: Newton's law of cooling, population growth, carbon dating.

Links to other subjects: Decay curves (physics); first order reactions (chemistry)

TOK: Does personal experience play a role in the formation of knowledge claims in mathematics? Does it play a different role in mathematics compared to other areas of knowledge?

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AHL 5.19

Content	Guidance, clarification and syllabus links
Maclaurin series to obtain expansions for e^x , $\sin x$, $\cos x$, $\ln(1+x)$, $(1+x)^p$, $p \in \mathbb{Q}$.	
Use of simple substitution, products, integration and differentiation to obtain other series.	Example: for substitution: replace x with x^2 to define the Maclaurin series for e^{x^2}. Example: the expansion of $e^x \sin x$.
Maclaurin series developed from differential equations.	

Connections

International-mindedness: Comparison of the Bourbaki to the Kerala School.

TOK: Is there always a trade-off between accuracy and simplicity?

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Assessment in the Diploma Programme

General

Assessment is an integral part of teaching and learning. The most important aim of assessment in the DP is that it should support curricular goals and encourage appropriate student learning. Both external and internal assessments are used in the DP. IB examiners mark work produced for external assessment, while work produced for internal assessment is marked by teachers and externally moderated by the IB.

There are two types of assessment identified by the IB.

Formative assessment informs both teaching and learning. It is concerned with providing accurate and helpful feedback to students and teachers on the kind of learning taking place and the nature of students' strengths and weaknesses in order to help develop students' understanding and capabilities. Formative assessment can also help to improve teaching quality, as it can provide information to monitor progress towards meeting the course aims and objectives.

Summative assessment gives an overview of previous learning and is concerned with measuring student achievement.

The DP primarily focuses on summative assessment designed to record student achievement at, or towards the end of, the course of study. However, many of the assessment instruments can also be used formatively during the course of teaching and learning, and teachers are encouraged to do this. A comprehensive assessment plan is viewed as being integral with teaching, learning and course organization. For further information, see the IB *Programme standards and practices* document.

The approach to assessment used by the IB is criterion-related, not norm-referenced. This approach to assessment judges students' work by their performance in relation to identified levels of attainment, and not in relation to the work of other students. For further information on assessment within the DP please refer to the publication *DP assessment: Principles and practice*.

To support teachers in the planning, delivery and assessment of the DP courses, a variety of resources can be found on the programme resource centre or purchased from the IB store (store.ibo.org). Additional publications such as specimen papers and markschemes, teacher support materials, subject reports and grade descriptors can also be found on the programme resource centre. Past examination papers as well as markschemes can be purchased from the IB store.

Methods of assessment

The IB uses several methods to assess work produced by students.

Assessment criteria

Assessment criteria are used when the assessment task is open-ended. Each criterion concentrates on a particular skill that students are expected to demonstrate. An assessment objective describes what students should be able to do, and assessment criteria describe how well they should be able to do it. Using assessment criteria allows discrimination between different answers and encourages a variety of responses. Each criterion comprises a set of hierarchically-ordered level descriptors. Each level descriptor is worth one or more marks. Each criterion is applied independently using a best-fit model. The maximum marks for each criterion may differ according to the criterion's importance. The marks awarded for each criterion are added together to give the total mark for the piece of work.

Markbands

Markbands are a comprehensive statement of expected performance against which responses are judged. They represent a single holistic criterion divided into level descriptors. Each level descriptor corresponds to a range of marks to differentiate student performance. A best-fit approach is used to ascertain which particular mark to use from the possible range for each level descriptor.

Analytic markschemes

Analytic markschemes are prepared for those examination questions that expect a particular kind of response and/or a given final answer from students. They give detailed instructions to examiners on how to break down the total mark for each question for different parts of the response.

Marking notes

For some assessment components marked using assessment criteria, marking notes are provided. Marking notes give guidance on how to apply assessment criteria to the particular requirements of a question.

Inclusive assessment arrangements

Inclusive assessment arrangements are available for candidates with assessment access requirements. These arrangements enable candidates with diverse needs to access the examinations and demonstrate their knowledge and understanding of the constructs being assessed.

The IB's *Access and inclusion policy* provides details on all the inclusive assessment arrangements available to candidates with learning support requirements. The IB document *Learning diversity and inclusion in the IB programmes* outlines the position of the IB with regard to candidates with diverse learning needs in the IB programmes. For candidates affected by adverse circumstances, the IB documents *General regulations: Diploma Programme* and *Diploma Programme Assessment procedures* provide details on access consideration.

Responsibilities of the school

The school is required to ensure that that equal access arrangements and reasonable adjustments are provided to candidates with learning support requirements that are in line with the IB documents *Access and inclusion policy* and *Learning diversity and inclusion in the IB programmes*.

Assessment outline—SL

First assessment 2021

Assessment component	Weighting
External assessment (3 hours) Paper 1 (90 minutes) No technology allowed. (80 marks) <i>Section A</i> Compulsory short-response questions based on the syllabus. <i>Section B</i> Compulsory extended-response questions based on the syllabus.	80% 40%
Paper 2 (90 minutes) Technology required. (80 marks) <i>Section A</i> Compulsory short-response questions based on the syllabus. <i>Section B</i> Compulsory extended-response questions based on the syllabus	40%
Internal assessment This component is internally assessed by the teacher and externally moderated by the IB at the end of the course. Mathematical exploration Internal assessment in mathematics is an individual exploration. This is a piece of written work that involves investigating an area of mathematics. (20 marks)	20%

Assessment outline—HL

First assessment 2021

Assessment component	Weighting
External assessment (5 hours) Paper 1 (120 minutes) No technology allowed. (110 marks) <i>Section A</i> Compulsory short-response questions based on the syllabus. <i>Section B</i> Compulsory extended-response questions based on the syllabus.	80% 30%
Paper 2 (120 minutes) Technology required. (110 marks) <i>Section A</i> Compulsory short-response questions based on the syllabus. <i>Section B</i> Compulsory extended-response questions based on the syllabus. Paper 3 (60 minutes) Technology required. (55 marks) Two compulsory extended response problem-solving questions.	30% 20%
Internal assessment This component is internally assessed by the teacher and externally moderated by the IB at the end of the course. Mathematical exploration Internal assessment in mathematics is an individual exploration. This is a piece of written work that involves investigating an area of mathematics. (20 marks)	20%

External assessment

General

Mark schemes are used to assess students in all papers. The mark schemes are specific to each examination.

External assessment details—SL

General information

Paper 1 and paper 2

These papers are externally set and externally marked. Together, they contribute 80% of the final mark for the course. These papers are designed to allow students to demonstrate what they know and what they can do.

Papers 1 and 2 will contain some questions, or parts of questions, which are common with HL.

Calculators

Paper 1

Students are not permitted access to any calculator. Questions will mainly involve analytic approaches to solutions, rather than requiring the use of a GDC. The paper is not intended to require complicated calculations, with the potential for careless errors. However, questions will include some arithmetical manipulations when they are essential to the development of the question.

Paper 2

Students must have access to a graphic display calculator (GDC) at all times. However, not all questions will necessarily require the use of the GDC. Regulations covering the types of GDC allowed are provided in Diploma Programme *Assessment procedures*.

Formula booklet

Each student must have access to a clean copy of the formula booklet during the examination. It is the responsibility of the school to download a copy from IBIS or the programme resource centre and to ensure that there are sufficient copies available for all students.

Awarding of marks

Marks are awarded for method, accuracy, answers and reasoning, including interpretation.

In paper 1 and paper 2, full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations (in the form of, for example diagrams, graphs or calculations). Where an answer is incorrect, some marks may be given for correct method, provided this is shown by written working. All students should therefore be advised to show their working.

Paper 1

Duration: 1 hour 30 minutes

Weighting: 40%

- This paper consists of section A, short-response questions, and section B, extended-response questions.

- Students are not permitted access to any calculator on this paper.

Syllabus coverage

- Knowledge of **all** topics is required for this paper. However, not all topics are necessarily assessed in every examination session.

Mark allocation

- This paper is worth **80** marks, representing **40%** of the final mark.
- Questions of varying levels of difficulty and length are set. Therefore, individual questions may not necessarily each be worth the same number of marks. The exact number of marks allocated to each question is indicated at the start of the question.

Section A

- This section consists of compulsory short-response questions based on the whole syllabus. It is worth approximately 40 marks.
- The intention of this section is to assess students across the breadth of the syllabus. However, it should not be assumed that the separate topics are given equal emphasis.

Question type

- A small number of steps are needed to solve each question.
- Questions may be presented in the form of words, symbols, diagrams or tables, or combinations of these.

Section B

- This section consists of a small number of compulsory extended-response questions based on the whole syllabus. It is worth approximately 40 marks.
- Individual questions may require knowledge of more than one topic.
- The intention of this section is to assess students across the breadth of the syllabus in depth. The range of syllabus topics tested in this section may be narrower than that tested in section A.

Question type

- Questions require extended responses involving sustained reasoning.
- Individual questions will develop a single theme.
- Questions may be presented in the form of words, symbols, diagrams or tables, or combinations of these.
- Normally, each question reflects an incline of difficulty, from relatively easy tasks at the start of a question to relatively difficult tasks at the end of a question. The emphasis is on sustained reasoning.

Paper 2

Duration: 1 hour 30 minutes

Weighting: 40%

- This paper consists of section A, short-response questions, and section B, extended-response questions.
- A GDC is required for this paper, but not every question will necessarily require its use.

Syllabus coverage

- Knowledge of **all** topics is required for this paper. However, not all topics are necessarily assessed in every examination session.

Mark allocation

- This paper is worth **80** marks, representing **40%** of the final mark.

- Questions of varying levels of difficulty and length are set. Therefore, individual questions may not necessarily each be worth the same number of marks. The exact number of marks allocated to each question is indicated at the start of the question.

Section A

- This section consists of compulsory short-response questions based on the whole syllabus. It is worth approximately 40 marks.
- The intention of this section is to assess students across the breadth of the syllabus. However, it should not be assumed that the separate topics are given equal emphasis.

Question type

- A small number of steps are needed to solve each question.
- Questions may be presented in the form of words, symbols, diagrams or tables, or combinations of these.

Section B

- This section consists of a small number of compulsory extended-response questions based on the whole syllabus. It is worth approximately 40 marks.
- Individual questions may require knowledge of more than one topic.
- The intention of this section is to assess students across the breadth of the syllabus in depth. The range of syllabus topics tested in this section may be narrower than that tested in section A.

Question type

- Questions require extended responses involving sustained reasoning.
- Individual questions will develop a single theme.
- Questions may be presented in the form of words, symbols, diagrams or tables, or combinations of these.
- Normally, each question reflects an incline of difficulty, from relatively easy tasks at the start of a question to relatively difficult tasks at the end of a question. The emphasis is on sustained reasoning.

General

Markschemes are used to assess students in all papers. The markschemes are specific to each examination.

External assessment details—HL

General information

Papers 1, 2 and 3

These papers are externally set and externally marked. Together, they contribute 80% of the final mark for the course. These papers are designed to allow students to demonstrate what they know and what they can do.

Papers 1 and 2 will contain some questions, or parts of questions, which are common with SL.

Calculators

Paper 1

Students are not permitted access to any calculator. Questions will mainly involve analytic approaches to solutions, rather than requiring the use of a GDC. The paper is not intended to require complicated calculations, with the potential for careless errors. However, questions will include some arithmetical manipulations when they are essential to the development of the question.

Paper 2

Students must have access to a GDC at all times. However, not all questions will necessarily require the use of the GDC. Regulations covering the types of GDC allowed are provided in *Diploma Programme Assessment procedures*.

Paper 3

Students must have access to a GDC at all times. However, not all question parts will necessarily require the use of the GDC. Regulations covering the types of GDC allowed are provided in *Diploma Programme Assessment procedures*.

Formula booklet

Each student must have access to a clean copy of the formula booklet during the examination. It is the responsibility of the school to download a copy from IBIS or the Programme Resource Centre and to ensure that there are sufficient copies available for all students.

Awarding of marks

Marks are awarded for method, accuracy, answers and reasoning, including interpretation.

In papers 1, 2 and 3, full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations (in the form of, for example diagrams, graphs or calculations). Where an answer is incorrect, some marks may be given for correct method, provided that this is shown by written working. All students should therefore be advised to show their working.

Paper 1

Duration: 2 hours

Weighting: 30%

- This paper consists of section A, short-response questions, and section B, extended-response questions.
- Students are not permitted access to any calculator on this paper.

Syllabus coverage

- Knowledge of **all** topics is required for this paper. However, not all topics are necessarily assessed in every examination session.

Mark allocation

- This paper is worth **110** marks, representing **30%** of the final mark.
- Questions of varying levels of difficulty and length are set. Therefore, individual questions may not necessarily each be worth the same number of marks. The exact number of marks allocated to each question is indicated at the start of the question.

Section A

- This section consists of compulsory short-response questions based on the whole syllabus. It is worth approximately 55 marks.
- The intention of this section is to assess students across the breadth of the syllabus. However, it should not be assumed that the separate topics are given equal emphasis.

Question type

- A small number of steps are needed to solve each question.
- Questions may be presented in the form of words, symbols, diagrams or tables, or combinations of these.

Section B

This section consists of a small number of compulsory extended-response questions based on the whole syllabus. It is worth approximately 55 marks.

Individual questions may require knowledge of more than one topic.

The intention of this section is to assess students across the breadth of the syllabus in depth. The range of syllabus topics tested in this section may be narrower than that tested in section A.

Question type

- Questions require extended responses.
- Individual questions will develop a single theme.
- Questions may be presented in the form of words, symbols, diagrams or tables, or combinations of these.
- Normally, each question reflects an incline of difficulty, from relatively easy tasks at the start of a question to relatively difficult tasks at the end of a question. The emphasis is on sustained reasoning.

Paper 2

Duration: 2 hours

Weighting: 30%

- This paper consists of section A, short-response questions, and section B, extended-response questions.
- A GDC is required for this paper, but not every question will necessarily require its use.

Syllabus coverage

- Knowledge of **all** topics is required for this paper. However, not all topics are necessarily assessed in every examination session.

Mark allocation

- This paper is worth **110** marks, representing **30%** of the final mark.
- Questions of varying levels of difficulty and length are set. Therefore, individual questions may not necessarily each be worth the same number of marks. The exact number of marks allocated to each question is indicated at the start of the question.

Section A

- This section consists of compulsory short-response questions based on the whole syllabus. It is worth approximately 55 marks.
- The intention of this section is to assess students across the breadth of the syllabus. However, it should not be assumed that the separate topics are given equal emphasis.

Question type

- A small number of steps are needed to solve each question.
- Questions may be presented in the form of words, symbols, diagrams or tables, or combinations of these.

Section B

- This section consists of a small number of compulsory extended-response questions based on the whole syllabus. It is worth approximately 55 marks.
- Individual questions may require knowledge of more than one topic.
- The intention of this section is to assess students across the breadth of the syllabus in depth. The range of syllabus topics tested in this section may be narrower than that tested in section A.

Question type

- Questions require extended responses.
- Individual questions will develop a single theme.
- Questions may be presented in the form of words, symbols, diagrams or tables, or combinations of these.
- Normally, each question reflects an incline of difficulty, from relatively easy tasks at the start of a question to relatively difficult tasks at the end of a question. The emphasis is on sustained reasoning.

Paper 3**Duration: 1 hour****Weighting: 20%**

- This paper consists of two compulsory extended-response problem-solving questions.
- A GDC is required for this paper, but not every question part will necessarily require its use.

Syllabus coverage

- Where possible, the first part of each question will be on syllabus content leading to the problem-solving context. Therefore, knowledge of all syllabus topics is required for this paper.

Mark allocation

- This paper is worth **55** marks, representing **20%** of the final mark.
- Questions may be unequal in terms of length and level of difficulty. Therefore, each question may not be worth the same number of marks. The exact number of marks allocated to each question is indicated at the start of each question.

Question type

- Questions require extended responses involving sustained reasoning.
- Individual questions will develop from a single theme where the emphasis is on problem solving leading to a generalization or the interpretation of a context.
- Questions may be presented in the form of words, symbols, diagrams or tables, or combinations of these.
- Normally, each question reflects an incline in difficulty, from relatively easy at the start of a question to relatively difficult tasks at the end of the question. The emphasis is on problem solving.

Internal assessment

Purpose of internal assessment

Internal assessment is an integral part of the course and is compulsory for both SL and HL students. It enables students to demonstrate the application of their skills and knowledge and to pursue their personal interests without the time limitations and other constraints that are associated with written examinations. The internal assessment should, as far as possible, be woven into normal classroom teaching and not be a separate activity conducted after a course has been taught.

The internal assessment requirements at SL and at HL is an individual exploration. This is a piece of written work that involves investigating an area of mathematics. It is marked according to five assessment criteria.

Guidance and authenticity

The exploration submitted for internal assessment must be the student's own work. However, it is not the intention that students should decide upon a title or topic and be left to work on the internal assessment component without any further support from the teacher. The teacher should play an important role during both the planning stage and the period when the student is working on the exploration.

It is the responsibility of the teacher to ensure that students are familiar with:

- the requirements of the type of work to be internally assessed
- the IB academic honesty policy available on the programme resource centre
- the assessment criteria; students must understand that the work submitted for assessment must address these criteria effectively.

Teachers and students must discuss the exploration. Students should be encouraged to initiate discussions with the teacher to obtain advice and information, and students must not be penalized for seeking guidance. As part of the learning process, teachers should read and give advice to students on one draft of the work. The teacher should provide oral or written advice on how the work could be improved, but not edit the draft. The next version handed to the teacher must be the final version for submission.

It is the responsibility of teachers to ensure that all students understand the basic meaning and significance of concepts that relate to academic honesty, especially authenticity and intellectual property. Teachers must ensure that all student work for assessment is prepared according to requirements and must explain clearly to students that the internally assessed work must be entirely their own.

All work submitted to the IB for moderation or assessment must be authenticated by a teacher, and must not include any known instances of suspected or confirmed malpractice. Each student must confirm that the work is his or her authentic work and constitutes the final version of that work. Once a student has officially submitted the final version of the work it cannot be retracted. The requirement to confirm the authenticity of work applies to the work of all students, not just the sample work that will be submitted to the IB for the purpose of moderation. For further details refer to the IB publications *Academic honesty in the IB educational context*, *The Diploma Programme: From principles into practice* and the relevant articles in *General regulations: Diploma Programme*.

Authenticity may be checked by discussion with the student on the content of the work, and scrutiny of one or more of the following:

- the student's initial proposal
- the draft of the written work
- the references cited
- the style of writing compared with work known to be that of the student

- the analysis of the work by a web-based plagiarism detection service such as www.turnitin.com.

The same piece of work cannot be submitted to meet the requirements of both the internal assessment and the extended essay.

Collaboration and teamwork

Collaboration and teamwork are a key focus of the approaches to teaching in the DP. It is advisable that the teacher uses the available class time to manage student collaboration. While working on their exploration students should be encouraged to work collaboratively in the various phases of the process, for example:

- generating ideas
- selecting the topic for their exploration
- sharing research sources
- acquiring the necessary knowledge, skills and understanding
- seeking peer feedback on their writing.

The approaches to teaching and learning (ATL) website on the programme resource centre provides an excellent source for developing collaborative skills in students.

While students should be encouraged to talk through their ideas with others, it is not appropriate to work together on a single exploration. It is important that students demonstrate how they incorporated sources and collaborative ideas into their work and that they always show their understanding and engagement in the work as described in the assessment criteria. Marks are awarded for the student's development and contribution to their exploration, not for work found in literature or carried out by others either individually or collaboratively.

It is imperative that students understand that the writing and calculations they do in their work must always be their own. This means that the argument they make and the ideas they rely on to make it, should either be their own or they should give credit to the source of those ideas. Any sources must be cited accordingly. This includes pictures, diagrams, graphs, formulae, etc.

In the specific cases of collecting information, data or measurements it is imperative that each student collects their own data, even in a case where collection of measurements is made from a group experiment. Group data or measurements can be combined to provide enough information for individual analysis and this should be clearly described in the written exploration.

Time allocation

Internal assessment is an integral part of the mathematics courses, contributing 20% to the final assessment in the SL and the HL courses. This weighting should be reflected in the time that is allocated to teaching the knowledge, skills and understanding required to undertake the work, as well as the total time allocated to carry out the work.

It is recommended that a total of approximately 10-15 hours of teaching time should be allocated to the work. This should include:

- time for the teacher to explain to students the requirements of the exploration
- class time for students to work on the exploration and ask questions
- time for consultation between the teacher and each student
- time to review and monitor progress, and to check authenticity.

Requirements and recommendations

Students can choose from a wide variety of activities: for example, modelling, investigations and applications of mathematics. To assist teachers and students in the choice of a topic, a list of stimuli is available in the teacher support material. However, students are not restricted to this list.

The exploration should be approximately 12-20 pages long with double line spacing, including diagrams and graphs, but excluding the bibliography. However, it is the quality of the mathematical writing that is important, not the length.

The teacher is expected to give appropriate guidance at all stages of the exploration by, for example, directing students into more productive routes of inquiry, making suggestions for suitable sources of information, and providing advice on the content and clarity of the exploration in the writing-up stage.

Teachers are responsible for indicating to students the existence of errors but should not explicitly correct these errors. It must be emphasized that students are expected to consult the teacher throughout the process.

All students should be familiar with the requirements of the exploration and the criteria by which it is assessed. Students need to start planning their explorations as early as possible in the course. Deadlines should be firmly established and adhered to. There should be a date for submission of the exploration topic and a brief outline description, a date for the submission of the draft and, of course, a date for completion.

In developing their explorations, students should aim to make use of mathematics learned as part of the course. The mathematics used should be commensurate with the level of the course—that is, it should be similar to that suggested in the syllabus. It is not expected that students produce work that is outside the syllabus—however, this will not be penalized.

Ethical guidelines should be adhered to throughout the planning and conducting of the exploration. Further details are given in the *Ethical practice in the Diploma Programme* poster on the programme resource centre.

Presentation

The following details should be stated on the cover page of the exploration:

- title of the exploration
- number of pages.

The references are not assessed. However, if they are not included in the final report it may be flagged in terms of academic honesty.

Using assessment criteria for internal assessment

For internal assessment, a number of assessment criteria have been identified. Each assessment criterion has level descriptors describing specific achievement levels, together with an appropriate range of marks. The level descriptors concentrate on positive achievement, although for the lower levels failure to achieve may be included in the description.

Teachers must judge the internally assessed work at SL and at HL against the criteria using the level descriptors.

The assessment criteria A to D are the same for both SL and HL. Criterion E “Use of mathematics” is different for SL and HL.

The aim is to find, for each criterion, the descriptor that conveys most accurately the level attained by the student, using the best-fit model. A best-fit approach means that compensation should be made when a piece of work matches different aspects of a criterion at different levels. The mark awarded should be one that most fairly reflects the balance of achievement against the criterion. It is not necessary for every single aspect of a level descriptor to be met for that mark to be awarded.

When assessing a student’s work, teachers should read the level descriptors for each criterion until they reach a descriptor that most appropriately describes the level of the work being assessed. If a piece of work seems to fall between two descriptors, both descriptors should be read again and the one that more appropriately describes the student’s work should be chosen.

Where there are two or more marks available within a level, teachers should award the upper marks if the student’s work demonstrates the qualities described to a great extent; the work may be close to achieving

marks in the level above. Teachers should award the lower marks if the student's work demonstrates the qualities described to a lesser extent; the work may be close to achieving marks in the level below.

Only whole numbers should be recorded; partial marks, (fractions or decimals) are not acceptable.

Teachers should not think in terms of a pass or fail boundary, but should concentrate on identifying the appropriate descriptor for each assessment criterion.

The highest-level descriptors do not imply faultless performance but should be achievable by a student. Teachers should not hesitate to use the extremes if they are appropriate descriptions of the work being assessed.

A student who attains a high achievement level in relation to one criterion will not necessarily attain high achievement levels in relation to the other criteria. Similarly, a student who attains a low achievement level for one criterion will not necessarily attain low achievement levels for the other criteria. Teachers should not assume that the overall assessment of the students will produce any particular distribution of marks.

It is recommended that the assessment criteria be made available to students.

Internal assessment details

Mathematical exploration

Duration: 10 to 15 hours

Weighting: 20%

Introduction

The internally-assessed component in this course is a mathematical exploration. This is a short report written by the student based on a topic chosen by him or her, and it should focus on the mathematics of that particular area. The emphasis is on mathematical communication (including formulae, diagrams, graphs, tables and so on), with his or her own focus, with the teacher providing feedback via, for example, discussion and interview. This will allow the students to develop areas of interest to them without a time constraint as in an examination, and allow all students to experience a feeling of success.

The final report should be approximately 12-20 pages long with double line spacing. It can be either word processed or handwritten. Students should be able to explain all stages of their work in such a way that demonstrates clear understanding. While there is no requirement that students present their work in class, it should be written in such a way that their peers would be able to follow it fairly easily. The report should include a detailed bibliography, and sources need to be referenced in line with the IB academic honesty policy. Direct quotes must be acknowledged.

The purpose of the exploration

The aims of the Mathematics: analysis and approaches and Mathematics: applications and interpretation courses at both SL and HL are carried through into the objectives that are formally assessed as part of the course, through either written examination papers or the exploration, or both. In addition to testing the objectives of the course, the exploration is intended to provide students with opportunities to increase their understanding of mathematical concepts and processes, and to develop a wider appreciation of mathematics. These are noted in the aims of the course. It is intended that, by doing the exploration, students benefit from the mathematical activities undertaken and find them both stimulating and rewarding. It will enable students to acquire the attributes of the IB learner profile.

The specific purposes of the exploration are to:

- develop students' personal insight into the nature of mathematics and to develop their ability to ask their own questions about mathematics
- provide opportunities for students to complete a piece of mathematical work over an extended period of time
- enable students to experience the satisfaction of applying mathematical processes independently

- provide students with the opportunity to experience for themselves the beauty, power and usefulness of mathematics
- encourage students, where appropriate, to discover, use and appreciate the power of technology as a mathematical tool
- enable students to develop the qualities of patience and persistence, and to reflect on the significance of their work
- provide opportunities for students to show, with confidence, how they have developed mathematically.

Management of the exploration

Work on the exploration should be incorporated into the course so that students are given the opportunity to learn the skills needed. Time in class can therefore be used for general discussion of areas of study, as well as familiarizing students with the criteria. Further details on the development of the exploration are included in the teacher support material.

Internal assessment criteria—SL and HL

The exploration is internally assessed by the teacher and externally moderated by the IB using assessment criteria that relate to the objectives for mathematics.

Each exploration is assessed against the following five criteria. The final mark for each exploration is the sum of the scores for each criterion. The maximum possible final mark is 20.

Students will not receive a grade for their mathematics course if they have not submitted an exploration.

Criterion A	Presentation
Criterion B	Mathematical communication
Criterion C	Personal engagement
Criterion D	Reflection
Criterion E	Use of mathematics

Criterion A: Presentation

Achievement level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	The exploration has some coherence or some organization.
2	The exploration has some coherence and shows some organization.
3	The exploration is coherent and well organized.
4	The exploration is coherent, well organized, and concise.

The “presentation” criterion assesses the organization and coherence of the exploration.

A **coherent** exploration is logically developed, easy to follow and meets its aim. This refers to the overall structure or framework, including introduction, body, conclusion and how well the different parts link to each other.

A **well-organized** exploration includes an introduction, describes the aim of the exploration and has a conclusion. Relevant graphs, tables and diagrams should accompany the work in the appropriate place and not be attached as appendices to the document. Appendices should be used to include information on large data sets, additional graphs, diagrams and tables.

A **concise** exploration does not show irrelevant or unnecessary repetitive calculations, graphs or descriptions.

The use of technology is not required but encouraged where appropriate. However, the use of analytic approaches rather than technological ones does not necessarily mean lack of conciseness, and should not be penalized. This does not mean that repetitive calculations are condoned.

Criterion B: Mathematical communication

Achievement level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	The exploration contains some relevant mathematical communication which is partially appropriate.
2	The exploration contains some relevant appropriate mathematical communication.
3	The mathematical communication is relevant, appropriate and is mostly consistent.
4	The mathematical communication is relevant, appropriate and consistent throughout.

The “mathematical communication” criterion assesses to what extent the student has:

- used appropriate mathematical language (**notation, symbols, terminology**). Calculator and computer notation is acceptable only if it is software generated. Otherwise it is expected that students use appropriate mathematical notation in their work
- defined **key terms** and variables, where required
- used **multiple forms of mathematical representation**, such as formulae, diagrams, tables, charts, graphs and models, where appropriate
- used a **deductive method** and set out proofs logically where appropriate

Examples of level 1 can include graphs not being labelled, consistent use of computer notation with no other forms of correct mathematical communication.

Level 4 can be achieved by using only one form of mathematical representation as long as this is appropriate to the topic being explored. For level 4, any *minor* errors that do not impair clear communication should not be penalized.

Criterion C: Personal engagement

Achievement level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	There is evidence of some personal engagement.
2	There is evidence of significant personal engagement.
3	There is evidence of outstanding personal engagement.

The “personal engagement” criterion assesses the extent to which the student engages with the topic by exploring the mathematics and making it their own. It is not a measure of effort.

Personal engagement may be recognized in different ways. These include thinking independently or creatively, presenting mathematical ideas in their own way, exploring the topic from different perspectives, making and testing predictions. Further (but not exhaustive) examples of personal engagement at different levels are given in the teacher support material (TSM).

There must be evidence of personal engagement demonstrated in the student's work. It is not sufficient that a teacher comments that a student was highly engaged.

Textbook style explorations or reproduction of readily available mathematics without the candidate's own perspective are unlikely to achieve the higher levels.

Significant: The student demonstrates authentic personal engagement in the exploration on a few occasions and it is evident that these drive the exploration forward and help the reader to better understand the writer's intentions.

Outstanding: The student demonstrates authentic personal engagement in the exploration in numerous instances and they are of a high quality. It is evident that these drive the exploration forward in a creative way. It leaves the impression that the student has developed, through their approach, a complete understanding of the context of the exploration topic and the reader better understands the writer's intentions.

Criterion D: Reflection

Achievement level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	There is evidence of limited reflection.
2	There is evidence of meaningful reflection.
3	There is substantial evidence of critical reflection.

The "reflection" criterion assesses how the student reviews, analyses and evaluates the exploration. Although reflection may be seen in the conclusion to the exploration, it may also be found throughout the exploration.

Simply describing results represents **limited reflection**. Further consideration is required to achieve the higher levels.

Some ways of showing **meaningful reflection** are: linking to the aims of the exploration, commenting on what they have learned, considering some limitation or comparing different mathematical approaches.

Critical reflection is reflection that is crucial, deciding or deeply insightful. It will often develop the exploration by addressing the mathematical results and their impact on the student's understanding of the topic. Some ways of showing critical reflection are: considering what next, discussing implications of results, discussing strengths and weaknesses of approaches, and considering different perspectives.

Substantial evidence means that the critical reflection is present throughout the exploration. If it appears at the end of the exploration it must be of high quality and demonstrate how it developed the exploration in order to achieve a level 3.

Further (but not exhaustive) examples of reflection at different levels are given in the teacher support material (TSM).

Criterion E: Use of mathematics—SL

Achievement level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	Some relevant mathematics is used.
2	Some relevant mathematics is used. Limited understanding is demonstrated.
3	Relevant mathematics commensurate with the level of the course is used. Limited understanding is demonstrated.

Achievement level	Descriptor
4	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is partially correct. Some knowledge and understanding are demonstrated.
5	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is mostly correct. Good knowledge and understanding are demonstrated.
6	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct. Thorough knowledge and understanding are demonstrated.

The “Use of mathematics” SL criterion assesses to what extent students use mathematics that is **relevant** to the exploration.

Relevant refers to mathematics that supports the development of the exploration towards the completion of its aim. Overly complicated mathematics where simple mathematics would suffice is not relevant.

Students are expected to produce work that is **commensurate with the level** of the course, which means it should not be completely based on mathematics listed in the prior learning. The mathematics explored should either be part of the syllabus, or at a similar level.

A key word in the descriptor is **demonstrated**. The command term demonstrate means “to make clear by reasoning or evidence, illustrating with examples or practical application”. Obtaining the correct answer is not sufficient to demonstrate understanding (even some understanding) in order to achieve level 2 or higher.

For knowledge and understanding to be **thorough** it must be demonstrated throughout.

The mathematics can be regarded as **correct** even if there are occasional minor errors as long as they do not detract from the flow of the mathematics or lead to an unreasonable outcome.

Students are encouraged to use technology to obtain results where appropriate, but **understanding must be demonstrated** in order for the student to achieve higher than level 1, for example merely substituting values into a formula does not necessarily demonstrate understanding of the results.

The mathematics only needs to be what is required to support the development of the exploration. This could be a few small elements of mathematics or even a single topic (or sub-topic) from the syllabus. It is better to do a few things well than a lot of things not so well. If the mathematics used is relevant to the topic being explored, commensurate with the level of the course and understood by the student, then it can achieve a high level in this criterion.

Criterion E: Use of mathematics—HL

Achievement level	Descriptor
0	The exploration does not reach the standard described by the descriptors below.
1	Some relevant mathematics is used. Limited understanding is demonstrated.
2	Some relevant mathematics is used. The mathematics explored is partially correct. Some knowledge and understanding is demonstrated.
3	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct. Some knowledge and understanding are demonstrated.

Achievement level	Descriptor
4	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct. Good knowledge and understanding are demonstrated.
5	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is correct and demonstrates sophistication or rigour. Thorough knowledge and understanding are demonstrated.
6	Relevant mathematics commensurate with the level of the course is used. The mathematics explored is precise and demonstrates sophistication and rigour. Thorough knowledge and understanding are demonstrated.

The “Use of mathematics” HL criterion assesses to what extent students use **relevant** mathematics in the exploration.

Students are expected to produce work that is **commensurate with the level** of the course, which means it should not be completely based on mathematics listed in the prior learning. The mathematics explored should either be part of the syllabus, at a similar level or slightly beyond. However, mathematics of a level slightly beyond the syllabus is **not** required to achieve the highest levels.

A key word in the descriptor is **demonstrated**. The command term demonstrate means to make clear by reasoning or evidence, illustrating with examples or practical application. Obtaining the correct answer is not sufficient to demonstrate understanding (even some understanding) in order to achieve level 2 or higher.

For knowledge and understanding to be **thorough** it must be demonstrated throughout. Lines of reasoning must be shown to justify steps in the mathematical development of the exploration.

Relevant refers to mathematics that supports the development of the exploration towards the completion of its aim. Overly complicated mathematics where simple mathematics would suffice is not relevant.

The mathematics can be regarded as **correct** even if there are occasional minor errors as long as they do not detract from the flow of the mathematics or lead to an unreasonable outcome. **Precise** mathematics is error-free and uses an appropriate level of accuracy at all times.

Sophistication: To be considered as sophisticated the mathematics used should be commensurate with the HL syllabus or, if contained in the SL syllabus, the mathematics has been used in a complex way that is beyond what could reasonably be expected of an SL student. Sophistication in mathematics may include understanding and using challenging mathematical concepts, looking at a problem from different perspectives and seeing underlying structures to link different areas of mathematics.

Rigour involves clarity of logic and language when making mathematical arguments and calculations. Mathematical claims relevant to the development of the exploration must be justified or proven.

Students are encouraged to use technology to obtain results where appropriate, but **understanding must be demonstrated** in order for the student to achieve level 1 or higher, for example merely substituting values into a formula does not necessarily demonstrate understanding of the results.

The mathematics only needs to be what is required to support the development of the exploration. This could be a few small elements of mathematics or even a single topic (or sub-topic) from the syllabus. It is better to do a few things well than a lot of things not so well. If the mathematics used is relevant to the topic being explored, commensurate with the level of the course and understood by the student, then it can achieve a high level in this criterion.

Glossary of command terms

Command terms for Mathematics: analysis and approaches

Students should be familiar with the following key terms and phrases used in examination questions, which are to be understood as described below. Although these terms will be used frequently in examination questions, other terms may be used to direct students to present an argument in a specific way.

Command term	Definition
Calculate	Obtain a numerical answer showing the relevant stages in the working.
Comment	Give a judgment based on a given statement or result of a calculation.
Compare	Give an account of the similarities between two (or more) items or situations, referring to both (all) of them throughout.
Compare and contrast	Give an account of similarities and differences between two (or more) items or situations, referring to both (all) of them throughout.
Construct	Display information in a diagrammatic or logical form.
Contrast	Give an account of the differences between two (or more) items or situations, referring to both (all) of them throughout.
Deduce	Reach a conclusion from the information given.
Demonstrate	Make clear by reasoning or evidence, illustrating with examples or practical application.
Describe	Give a detailed account.
Determine	Obtain the only possible answer.
Differentiate	Obtain the derivative of a function.
Distinguish	Make clear the differences between two or more concepts or items.
Draw	Represent by means of a labelled, accurate diagram or graph, using a pencil. A ruler (straight edge) should be used for straight lines. Diagrams should be drawn to scale. Graphs should have points correctly plotted (if appropriate) and joined in a straight line or smooth curve.
Estimate	Obtain an approximate value.
Explain	Give a detailed account including reasons or causes.
Find	Obtain an answer showing relevant stages in the working.
Hence	Use the preceding work to obtain the required result.
Hence or otherwise	It is suggested that the preceding work is used, but other methods could also receive credit.
Identify	Provide an answer from a number of possibilities.
Integrate	Obtain the integral of a function.

Command term	Definition
Interpret	Use knowledge and understanding to recognize trends and draw conclusions from given information.
Investigate	Observe, study, or make a detailed and systematic examination, in order to establish facts and reach new conclusions.
Justify	Give valid reasons or evidence to support an answer or conclusion.
Label	Add labels to a diagram.
List	Give a sequence of brief answers with no explanation.
Plot	Mark the position of points on a diagram.
Predict	Give an expected result.
Prove	Use a sequence of logical steps to obtain the required result in a formal way.
Show	Give the steps in a calculation or derivation.
Show that	Obtain the required result (possibly using information given) without the formality of proof. "Show that" questions do not generally require the use of a calculator.
Sketch	Represent by means of a diagram or graph (labelled as appropriate). The sketch should give a general idea of the required shape or relationship, and should include relevant features.
Solve	Obtain the answer(s) using algebraic and/or numerical and/or graphical methods.
State	Give a specific name, value or other brief answer without explanation or calculation.
Suggest	Propose a solution, hypothesis or other possible answer.
Verify	Provide evidence that validates the result.
Write down	Obtain the answer(s), usually by extracting information. Little or no calculation is required. Working does not need to be shown.

Notation list

There are various systems of notation in use, and the IB has chosen to adopt a system based on the recommendations of the International Organization for Standardization (ISO). This notation is used in the examination papers for this course without explanation. If forms of notation other than those listed in this guide are used on a particular examination paper, they are defined within the question in which they appear.

Because students are required to recognize, though not necessarily use, IB notation in examinations, it is recommended that teachers introduce students to this notation at the earliest opportunity. Students are **not** allowed access to information about this notation in the examinations.

Students must always use correct mathematical notation, not calculator notation.

SL and HL

$\mathbb{N}$	the set of positive integers and zero, $\{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$	the set of integers, $\{0, \pm 1, \pm 2, \pm 3, \dots\}$
$\mathbb{Z}^+$	the set of positive integers, $\{1, 2, 3, \dots\}$
$\mathbb{Q}$	the set of rational numbers
$\mathbb{Q}^+$	the set of positive rational numbers, $\{x \mid x \in \mathbb{Q}, x > 0\}$
$\mathbb{R}$	the set of real numbers
$\mathbb{R}^+$	the set of positive real numbers, $\{x \mid x \in \mathbb{R}, x > 0\}$
$\{x_1, x_2, \dots\}$	the set with elements $x_1, x_2, \dots$
$n(A)$	the number of elements in the finite set A
$\{x \mid \quad \}$	the set of all x such that
$\in$	is an element of
$\notin$	is not an element of
$\emptyset$	the empty (null) set
U	the universal set
$\cup$	union
$\cap$	intersection
A'	the complement of the set A
$a^{1/2}, \sqrt{a}$	a to the power $\frac{1}{2}$, square root of a (if $a \geq 0$ then $\sqrt{a} \geq 0$)
$a^{1/n}, \sqrt[n]{a}$	a to the power of $\frac{1}{n}$, n^{th} root of a (if $a \geq 0$ then $\sqrt[n]{a} \geq 0$)
$a^{-n} = \frac{1}{a^n}$	a to the power of $-n$, reciprocal of a^n
$ x $	the modulus or absolute value of x , that is $\begin{cases} x & \text{for } x \geq 0, x \in \mathbb{R} \\ -x & \text{for } x < 0, x \in \mathbb{R} \end{cases}$

$\equiv$	identity
$\approx$	is approximately equal to
$>$	is greater than
$\geq$	is greater than or equal to
$<$	is less than
$\leq$	is less than or equal to
$\nlessgtr$	is not greater than
$\nlessgtr$	is not less than
$\Rightarrow$	implies
$\Leftrightarrow$	implies and is implied by
u_n	the n^{th} term of a sequence or series
d	the common difference of an arithmetic sequence
r	the common ratio of a geometric sequence
S_n	the sum of the first n terms of a sequence, $u_1 + u_2 + \dots + u_n$
S_∞	the sum to infinity of a sequence, $u_1 + u_2 + \dots$
$\sum_{i=1}^n u_i$	$u_1 + u_2 + \dots + u_n$
$n!$	$n(n-1)(n-2)\dots 3 \times 2 \times 1$
nC_r	$\frac{n!}{r!(n-r)!}$
Δ	the discriminant of a quadratic equation, $\Delta = b^2 - 4ac$
$f(x)$	the image of x under the function f
f^{-1}	the inverse function of the function f
$f \circ g$	the composite function of f and g
$\frac{dy}{dx}$	the derivative of y with respect to x
$f'(x)$	the derivative of $f(x)$ with respect to x
$\frac{d^2y}{dx^2}$	the second derivative of y with respect to x
$f''(x)$	the second derivative of $f(x)$ with respect to x
$\int y dx$	the indefinite integral of y with respect to x
$\int_a^b y dx$	the definite integral of y with respect to x between the limits $x = a$ and $x = b$
e^x	the exponential function of x
$\log_a x$	the logarithm to the base a of x
$\ln x$	the natural logarithm of x , $\log_e x$
$\sin, \cos, \tan$	the circular functions
$A(x, y)$	the point A in the plane with Cartesian coordinates x and y

$[AB]$	the line segment with end points A and B
AB	the length of $[AB]$
(AB)	the line containing points A and B
$\hat{A}$	the angle at A
$\hat{CAB}$	the angle between $[CA]$ and $[AB]$
ΔABC	the triangle whose vertices are A, B and C
$P(A)$	probability of event A
$P(A')$	probability of the event "not A"
$P(A B)$	probability of the event A given B
$x_1, x_2, \dots$	observations
$f_1, f_2, \dots$	frequencies with which the observations $x_1, x_2, \dots$ occur
$E(X)$	the expected value of the random variable X
μ	population mean
σ^2	population variance
σ	population standard deviation
$\bar{x}$	the sample mean of a set $\{x_1, x_2, \dots, x_n\}$ of observations
$P(X = x)$	the probability that the random variable X takes the value x
$B(n, p)$	binomial distribution with parameters n and p
$N(\mu, \sigma^2)$	normal distribution with mean μ and variance σ^2
$X \sim B(n, p)$	the random variable X has a binomial distribution with parameters n and p
$X \sim N(\mu, \sigma^2)$	the random variable X has a normal distribution with mean μ and variance σ^2
r	Pearson's product-moment correlation coefficient

AHL only

$\mathbb{C}$	the set of complex numbers, $\{a + bi \mid a, b \in \mathbb{R}\}$
i	$\sqrt{-1}$ where $i^2 = -1$
z	a complex number
z^*	the complex conjugate of z
$ z $	the modulus of z
$\arg z$	the argument of z
$\operatorname{Re} z$	the real part of z
$\operatorname{Im} z$	the imaginary part of z
$\operatorname{cis} \theta$	$\cos \theta + i \sin \theta$
$e^{i\theta}$	Euler/exponential form of a complex number

${}^n\text{P}_r$	$\frac{n!}{(n-r)!}$
$\Leftarrow$	is implied by
$[a, b]$	the closed interval $a \leq x \leq b$
$]a, b[$	the open interval $a < x < b$
$f: A \rightarrow B$	f is a function under which each element of a set A has an image in set B .
$\lim_{x \rightarrow a} f(x)$	the limit of $f(x)$ as x tends to a
$\frac{d^n y}{dx^n}$	the n^{th} derivative of y with respect to x
$f^{(n)}(x)$	the n^{th} derivative of $f(x)$ with respect to x
$\left. \begin{array}{l} \arcsin, \sin^{-1} \\ \arccos, \cos^{-1} \\ \arctan, \tan^{-1} \end{array} \right\}$	the inverse circular functions
cosec, sec, cot	the reciprocal circular functions
$\mathbf{v}$	the vector $\mathbf{v}$
$\overrightarrow{AB}$	the vector represented in magnitude and direction by the directed line segment from A to B
$\mathbf{a}$	the position vector $\overrightarrow{OA}$
$\mathbf{i}, \mathbf{j}, \mathbf{k}$	unit vectors in the directions of the Cartesian coordinate axes
$ \mathbf{a} $	the magnitude of $\mathbf{a}$
$ \overrightarrow{AB} $	the magnitude of $\overrightarrow{AB}$
$\mathbf{v} \cdot \mathbf{w}$	the scalar product of $\mathbf{v}$ and $\mathbf{w}$
$\mathbf{v} \times \mathbf{w}$	the vector product of $\mathbf{v}$ and $\mathbf{w}$
$f(x)$	the probability density function of the continuous random variable X
$\text{Var}(X)$	the variance of the random variable X